

Vangl2 suppresses NF- κ B signaling and ameliorates sepsis by targeting p65 for NDP52-mediated autophagic degradation

Reviewed Preprint

v2 • July 3, 2024

Revised by authors

Reviewed Preprint

v1 • May 12, 2023

Jiansen Lu, Jiahuan Zhang, Huaji Jiang, Zhiqiang Hu, Yufen Zhang, Lian He, Jianwu Yang, Yingchao Xie, Dan Wu, Hongyu Li, Ke Zeng, Peng Tan, Qingyue Xiao, Zijing Song, Chenglong Pan, Xiaochun Bai , Xiao Yu 

Department of Joint Surgery, the Fifth Affiliated Hospital, Southern Medical University, Guangzhou, Guangdong 510515, China • Department of Immunology, School of Basic Medical Sciences, Southern Medical University, Guangzhou, Guangdong 510515, China • Laboratory Medicine, Guangdong Provincial People's Hospital (Guangdong Academy of Medical Sciences), Southern Medical University, Guangzhou, Guangdong 510515, China • Department of Orthopaedics, Yue Bei People's Hospital Affiliated to Medical College of Shantou University, Shaoguan 512026, China • Department of Pharmacology, School of Medicine, Southern University of Science and Technology, Shenzhen, Guangdong 518005, China • Institute of Biosciences and Technology, College of Medicine, Texas A&M University, Houston, Texas 77030, USA • Klarman Cell Observatory, Broad Institute of MIT and Harvard, Cambridge, Massachusetts 02142, USA • Department of Cell Biology, School of Basic Medical Sciences, Southern Medical University, Guangzhou, Guangdong 510515, China • Guangdong Provincial Key Lab of Single Cell Technology and Application, Southern Medical University, Guangzhou, Guangdong 510515, China

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

Abstract

Van Gogh-like 2 (Vangl2), a core planar cell polarity (PCP) component, plays an important role in polarized cellular and tissue morphology induction, growth development and cancer. However, its role in regulating inflammatory responses remains elusive. Here, we report that Vangl2 is upregulated in patients with sepsis and identify Vangl2 as a negative regulator of NF- κ B signaling by regulating the protein stability and activation of the core transcription component p65. Mice with myeloid-specific deletion of Vangl2 (*Vangl2^{ΔM}*) are hypersusceptible to lipopolysaccharide (LPS)-induced septic shock. Vangl2 deficient myeloid cells exhibit enhanced phosphorylation and expression of p65, therefore, promoting the secretion of pro-inflammatory cytokines after LPS stimulation. Mechanistically, NF- κ B signaling-induced-Vangl2 recruits E3 ubiquitin ligase PDLIM2 to catalyze K63-linked ubiquitination on p65, which serves as a recognition signal for cargo receptor NDP52-mediated selective autophagic degradation. Taken together, these findings demonstrate Vangl2 as a suppressor of NF- κ B mediated inflammation and provide insights into the crosstalk between autophagy and inflammatory diseases.

eLife assessment

This **valuable** manuscript describes a novel role of Vangl2, a core planar cell polarity protein, in linking the NF- κ B pathway to selective autophagic protein degradation in myeloid cells. The mechanistic studies suggest that Vangl2 targets p65 for NDP52-mediated autophagic degradation, limiting inflammatory NF- κ B response, with functional significance of the proposed mechanism in sepsis. The presented evidence is **convincing**. Additional studies dissecting autophagic Vangl2 functions in various myeloid subsets in the context of inflammation could be informative, and additional Vangl2 targets in the inflammatory pathway, including IKK2, could also be explored. Overall, this exciting study will likely advance our understanding of NF- κ B control, particularly in the context of inflammatory diseases.

<https://doi.org/10.7554/eLife.87935.2.sa3>

Introduction

Vangl2, a core PCP component, mediates Wntless-type (Wnt)/PCP signaling, and controls homeostasis, development, and repair of organs (Bailly, Walton, & Borg, 2018; Brunt et al., 2021; Hatakeyama, Wald, Printsev, Ho, & Carraway, 2014). Vangl2 has four transmembrane domains with both carboxyl termini and amino oriented toward the cytoplasm, which is phosphorylated in the endoplasmic reticulum (ER) and then transported to the cell surface and becomes stabilized, while the unphosphorylated Vangl2 is unstable and internalized to degrade via the lysosomal pathway (D. Feng et al., 2021). The shuttle of Vangl2 between cytoplasm and cell membrane results in its multifunction, including adhesion, membrane protrusive activity, migration, and bridging proteins (Hatakeyama et al., 2014). Indeed, Vangl2 inhibited matrix metalloproteinase 2 (MMP2) activity and affected cell adhesion to extracellular matrix proteins (Jessen & Jessen, 2017). Vangl2 also modulates glomerular injury by promoting MMP9 (Papakrivopoulou et al., 2018). Moreover, the abnormal function of Vangl2 results in various diseases, such as cancer, kidney glomerular injury, idiopathic pulmonary fibrosis and systemic dysplasia (Papakrivopoulou et al., 2018; Poobalasingam et al., 2017). Vangl2 is markedly down-regulated in patients with emphysema (Poobalasingam et al., 2017), while which level was up-regulated and amplified in breast, ovarian, and uterine carcinomas (Cancer Genome Atlas Research et al., 2013; Cerami et al., 2012; Gao et al., 2013). In addition, Vangl2 is bounded to p62 to promote breast cancer (Puvirajesinghe et al., 2016). Our previous study showed that Vangl2 prevents osteogenic differentiation in mesenchymal stem cells, resulting in osteogenic dysplasia (Gong et al., 2021). Vangl2 mediated downstream factors of Toll-like or interleukin (IL)-1 receptor, such as myeloid differentiation factor 88 (MyD88) (Gong et al., 2021), suggesting that Vangl2 may play roles in immune-related diseases, including autoimmune diseases. However, the function of Vangl2 in inflammatory diseases remains uncovered.

The NF- κ B signaling is critical for the pathogenesis of a number of inflammatory diseases, such as inflammatory bowel disease (IBD), rheumatoid arthritis (RA), systemic lupus erythematosus (SLE) and sepsis (Liu, Zhang, Joo, & Sun, 2017). NF- κ B activation relies on the pattern recognition receptors (PRRs) recognition of pathogen-associated molecular patterns (PAMPs), including Toll like receptor (TLR), NOD-like receptors (NLRs) and RIG-I-like receptors (RLRs) (Liu et al., 2017). LPS triggers TLR activation and the recruitment of adaptor proteins including MyD88. This in turn activates a series of downstream canonical NF- κ B signaling cascade, resulting in the phosphorylation and degradation of I κ B and the nuclear translocation of RelA/p65 and p50 to induce the transcription of several inflammatory cytokines, such as interleukin (IL)-1, IL-6, and

tumor necrosis factor- α (TNF- α) (Funes, Rios, Escobar-Vera, & Kalergis, 2018 [↗](#); Locati, Curtale, & Mantovani, 2020 [↗](#)). Since uncontrolled immune responses are detrimental to the host, inappropriate or excessive NF- κ B activity contributes to the pathogenesis of various inflammatory diseases and cancer (Cartwright, Perkins, & C, 2016 [↗](#)). Thus NF- κ B signaling must be tightly regulated to maintain immune balance in the organism. In recent decades, extensive studies have focused on the mechanisms underlying the regulation of NF- κ B signaling. Recent research demonstrated that NLRC5 strongly prevents NF- κ B signaling pathway by interacting with I κ B kinase (IKK) α /IKK β and blocking their phosphorylation (Cui et al., 2010 [↗](#)). COMMD1, PPAR γ , SOCS1 and GCN5 were also shown to negatively regulate NF- κ B signaling (Bartuzi, Hofker, & van de Sluis, 2013 [↗](#); Mao et al., 2009 [↗](#)). Meanwhile, tripartite motif-containing protein 21 (Trim21) enhanced the interaction of p65 with IKK, which promotes p65 phosphorylation and downstream gene activation (Yang et al., 2021 [↗](#)). However, the molecular mechanisms underpinning the regulation of NF- κ B signaling are still elusive.

Autophagy, a conserved intracellular degradation pathway, decomposes cytoplasmic organelles and components, and acts as a defense mechanism to pathogen infection, playing a crucial role in nutrient recycling, stress response and cellular homeostasis (Ashrafi & Schwarz, 2013 [↗](#); Denton & Kumar, 2019 [↗](#)). Recent evidence indicates that autophagy is highly selective when delivering specific substrates to autolysosomal degradation by virtue of a number of cargo receptors, including sequestosome 1 (SQSTM1/p62), optineurin (OPTN), nuclear dot protein 52 (NDP52/CALCOCO2) and neighbor of BRCA1 (NBR1) (Gatica, Lahiri, & Klionsky, 2018 [↗](#)). Selective autophagy targets immune regulators for degradation, thus suppressing innate immune signaling, such as type I IFN and NF- κ B signaling (Pradel, Robert-Hebmann, & Espert, 2020 [↗](#); Tong et al., 2012 [↗](#)). Moreover, p62 protein has been identified as a novel Vangl2-binding partner (Puvirajesinghe et al., 2016 [↗](#)), and our recent study has demonstrated that Vangl2 reduces lysosomal chaperone-mediated autophagy (CMA) activity by targeting LAMP-2A for degradation (Gong et al., 2021 [↗](#)). However, whether and how Vangl2 is involved in the selective autophagic regulation of NF- κ B signaling remains largely unknown.

In this study, we uncover a previously unrecognized role of Vangl2, as a “molecular brake”, in the negative regulation of NF- κ B signaling to prevent excessive and potentially harmful immune responses during sepsis in both human patient samples and LPS induced mouse model. Induction of Vangl2 upon inflammation recruits an E3 ubiquitin ligase PDLIM2 to catalyze K63-linked ubiquitination on p65, thus promoting the recognition of p65 by the cargo receptor NDP52 and resulting in the selective autophagic degradation of p65. Our findings provide a potential target for the treatment of inflammatory diseases.

Results

Loss of Vangl2 promotes inflammatory responses in LPS-induced septic shock

To investigate the potential role of Vangl2 in inflammatory response, we first analyzed the expression of Vangl2 in peripheral blood mononuclear cells (PBMCs) from sepsis patients (Fig. S1 A) and found mRNA level of Vangl2 was increased in the sepsis patients compared to healthy control (HC) (Fig. 1 A [↗](#)). Moreover, the serum of the high Vangl2 expression group had lower levels of IL-6, white blood cell (WBC), and acute C-reactive protein (CRP) than the low Vangl2 expression group (Fig. S1 A). Expression of Vangl2 from database GSE156382 also confirmed that Vangl2 mRNA was induced during sepsis (Fig. S1 B). To further determine whether *Vangl2* expression could be regulated in response to inflammatory stimulation, we treated mice with LPS to activate NF- κ B pathway and detected *Vangl2* mRNA in different tissues from LPS treated mice, and found the expression of *Vangl2* significantly increased in secondary lymphoid organs including the spleen and lymph nodes after LPS stimulation, but not in other tissues (Fig. 1 B [↗](#)).

Moreover, qPCR and western blot analyses revealed strong upregulation of *Vangl2* at both the mRNA and protein levels after LPS stimulation in bone marrow (BM)-derived macrophages (BMDMs), neutrophils and peritoneal macrophages (pMACs) (Fig. S1, C-E), suggesting that *Vangl2* is induced in immune organ tissues and cells in response to inflammation.

To determine the function of *Vangl2* during LPS-induced sepsis, we specifically ablated *Vangl2* in myeloid cells by crossing *Vangl2^{flox/flox}* mice with mice expressing lysozyme proximal promoter (*Lyz2-Cre*). The resultant homozygous *Vangl2^{flox/flox} × Lyz2-Cre* mice were designated *Vangl2^{ΔM}* mice and selective deletion of *Vangl2* in myeloid cells of *Vangl2^{ΔM}* mice were confirmed by PCR (Fig. S1 F). Although there were no significant differences in the size of spleen and lymph nodes between wild-type (WT) and *Vangl2^{ΔM}* mice (Fig. S1 G), myeloid-specific loss of *Vangl2* increased the number of monocytes, macrophages and neutrophils in the spleen and bone marrow (Fig. S1, H and I). To gain further insight into the physiological function of *Vangl2*, we treated WT and *Vangl2^{ΔM}* mice with a high dose of LPS and monitored mice survival. We found that all *Vangl2^{ΔM}* mice died after LPS-induced septic shock within 20 hours, compared to only 20% of the WT mice, whereas the remaining WT mice survived for more than 50 hours (Fig. 1 C). Consistent with this observation, *Vangl2^{ΔM}* mice markedly increased the protein level of IL-1β in isolated CD11b⁺ splenocytes (Fig. 1 D), and mRNA levels of *Il1b*, *Tnfa* and *Il6* in the spleen (Fig. 1 E). Meanwhile, *Vangl2^{ΔM}* mice showed markedly elevated serum amounts of proinflammatory cytokines such as IL-1β, TNF-α and IL-6 after LPS treatment, compared with WT mice (Fig. 1 F). Together, these data provide in vivo evidence that myeloid cell-specific deletion of *Vangl2* in mice enhances the sensitivity and severity of LPS-induced septic shock and is associated with increased expression of proinflammatory cytokines.

Vangl2 negatively regulates NF-κB activation and inflammation in myeloid cells

To investigate the mechanisms by which *Vangl2* prevents sepsis, we performed RNA-seq analysis to identify signal pathways involved in LPS-induced septic shock by comparing LPS-stimulated BMDMs from *Vangl2^{ΔM}* and WT mice. Results showed 907 up-regulated genes (including inflammatory cytokines) and 1092 down-regulated genes in response to *Vangl2* deficiency in BMDMs after LPS stimulation (Fig. S2 A). Consistent with Fig. 1, gene ontology (GO) analysis further identified that these genes are involved in cellular immune responses, including “cellular response to lipopolysaccharide” (Fig. S2 B). Moreover, KEGG analysis revealed genes involved in TNF signaling pathway and cytokine-cytokine receptor interaction were highly enriched in *Vangl2^{ΔM}* BMDMs after LPS stimulation, suggesting that *Vangl2* may regulate these signaling pathways and related cytokines release (Fig. S2 C).

To determine the function of *Vangl2* in innate immune signaling in myeloid cells, we isolated BMDMs, neutrophils and pMACs from WT and *Vangl2^{ΔM}* mice, treated them with LPS, and performed immunoblot analysis with specific antibodies. We found that phosphorylations of p65 were enhanced and remained high level for a sustained period in *Vangl2*-deficient pMACs and neutrophils, compared with WT cells (Fig. 2, A-D). Consistent with this observation, *Vangl2*-deficient macrophages, and neutrophils showed markedly elevated pro-inflammatory cytokines such as TNF-α, IL-6 (Fig. 2 E and F and Fig. S2 D and E) and IL-1β (Fig. S2 F), after LPS treatment. Moreover, we detected enhanced p65 nuclear accumulation in LPS-induced *Vangl2*-deficient neutrophils, compared with WT neutrophils (Fig. 2, G and H).

To further confirm the function of *Vangl2* in regulating NF-κB signaling, we overexpressed Flag-tagged *Vangl2* in A549 cells. We found that *Vangl2* overexpression inhibited phosphorylation of p65 in A549 cells (Fig. 2, I and J). These results suggest that *Vangl2* prevents LPS-induced NF-κB activation and proinflammatory cytokine production.

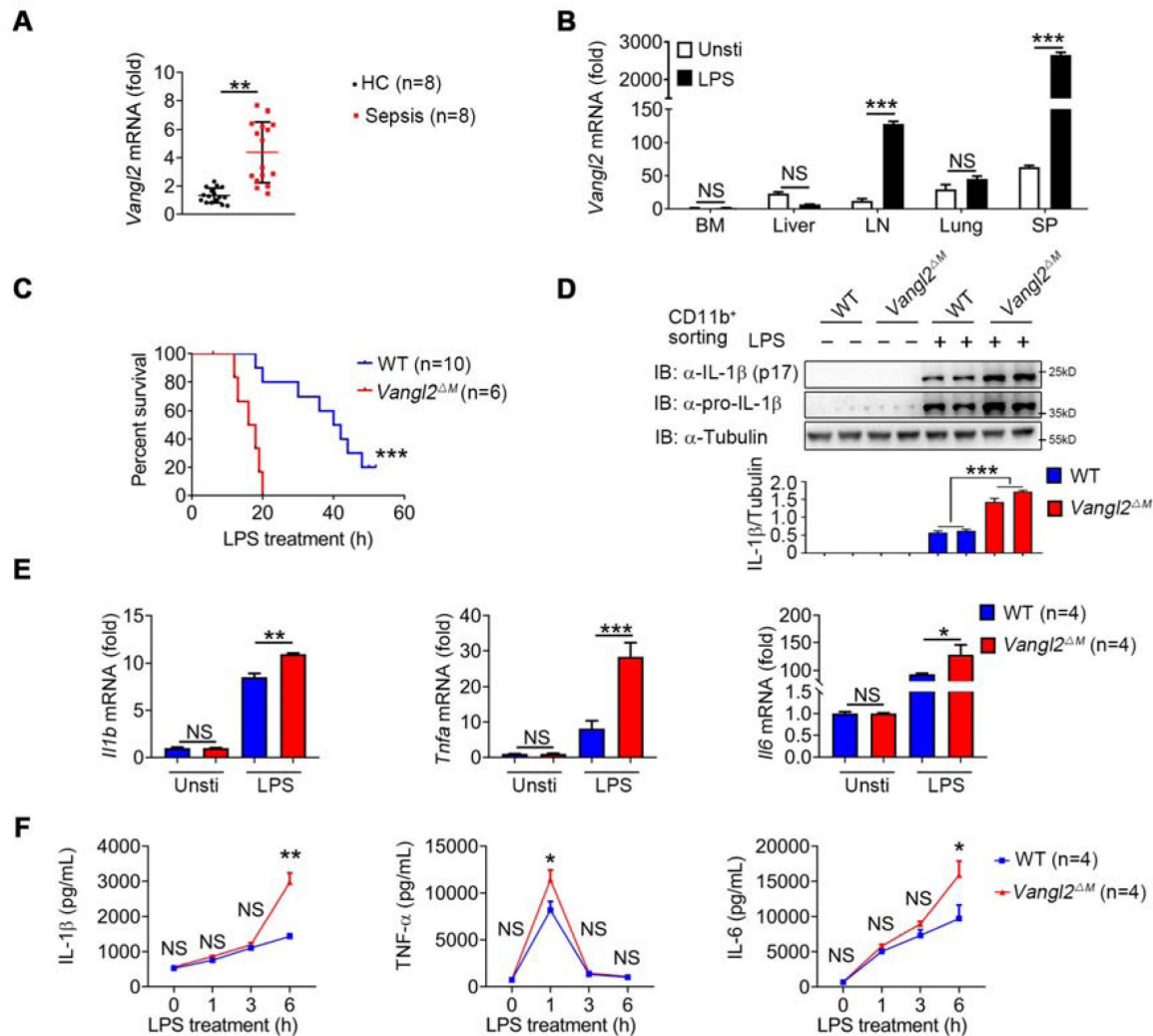


Figure 1.

Vangl2 ablation promotes inflammation during LPS treatment.

(A) Transcription levels of *Vangl2* in PBMCs from healthy volunteers (healthy control, HC) and sepsis patients were analyzed by real-time PCR (n=8).

(B) *Vangl2* mRNA in different organs from mice treated with or without LPS (n≥3).

(C) The survival rates of WT and *Vangl2*^{ΔM} mice treated with high-dosage of LPS (30 mg/kg, i.p.) (n≥4).

(D-F) WT and *Vangl2*^{ΔM} mice (n≥3) were treated with LPS (30 mg/kg, i.p.). Splenocytes were collected at 9 h after LPS treatment. Cell lysates of CD11b⁺ sorted splenocytes were analyzed by immunoblotting with the indicated antibodies **(D)**.

RNAs from splenocytes were isolated and used for expression analysis of *Il1b*, *Tnfa*, and *Il6* using qPCR **(E)**. Sera were collected at indicated times post LPS treatment and subjected to ELISA analysis of IL-1β, TNF-α, and IL-6 **(F)**. PBMCs, peripheral blood mononuclear cells; Unsti, unstimulation; LPS, lipopolysaccharide; LN, lymph node; SP, spleen. Data are representative of three independent experiments and are plotted as the mean ±SD. **p*<0.05, ***p*<0.01, ****p*<0.001 vs. corresponding control.

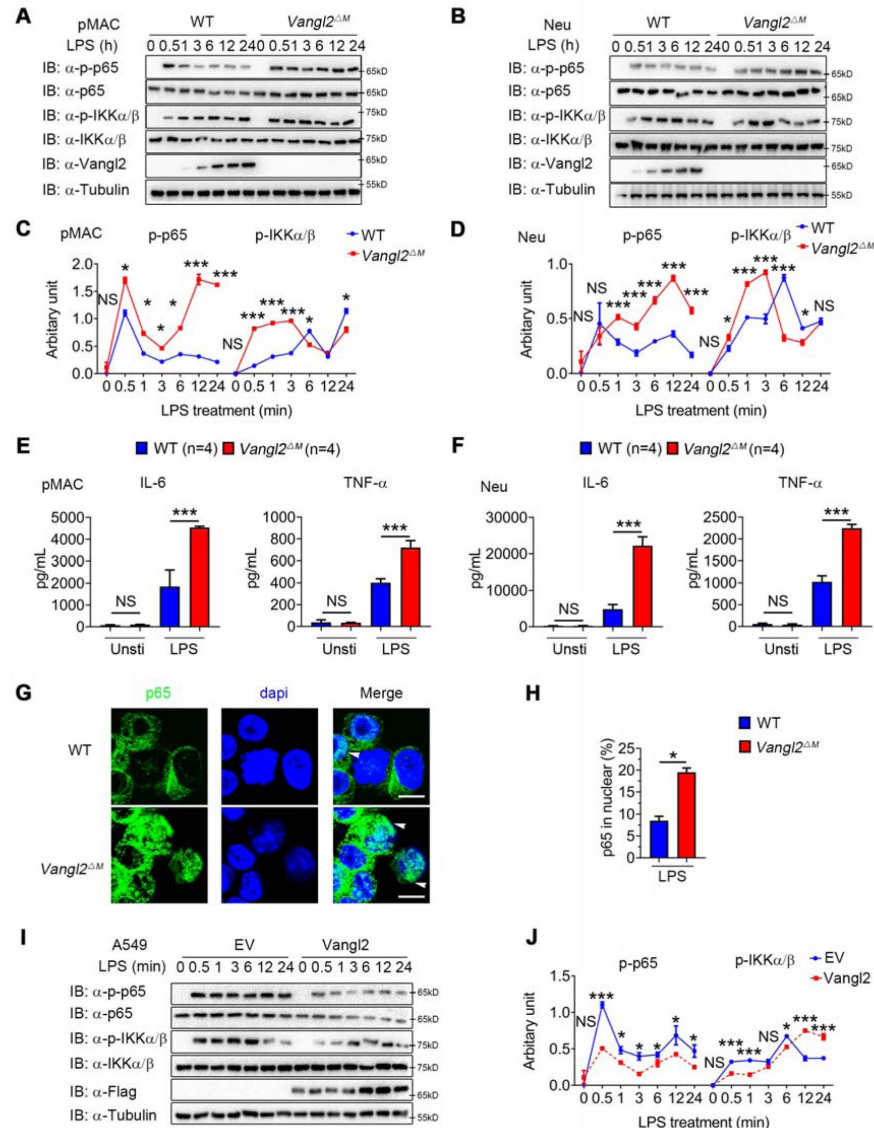


Figure 2.

Vangl2 negatively regulates LPS-induced NF- κ B activation and proinflammatory cytokines.

(A-D) WT and Vangl2-deficient ($n \geq 3$) pMAC (A and C) or neutrophils (B and D) were stimulated with LPS (100 ng/ml) for the indicated times. Immunoblot analysis of total and phosphorylated p65, IKK α/β (A and B), and analysis of grey intensity was shown (C and D).

(E-F) WT and Vangl2-deficient ($n \geq 4$) pMAC or neutrophils were stimulated with LPS (100 ng/ml) for 6 h. mRNA levels of *Il6* and *Tnfa* were measured by qPCR (E). IL-6 and TNF- α secretion by WT and Vangl2-deficient BMDMs or neutrophils treated with or without LPS for 6 h was measured by ELISA (F).

(G and H) The WT and Vangl2-deficient ($n \geq 3$) neutrophils were treated with LPS (1000 ng/ml) for 4 h, and the nuclear translocation of p65 was detected by immunofluorescence (G) (p65, green; DAPI, blue; Scale bar, 50 μ m). Percentages of p65 nuclear translocated cells in WT and Vangl2-deficient neutrophils were determined by counting 100 to 150 cells in non-overlapping fields (H).

(I and J) A549 cells were transfected with Flag-tagged Vangl2 plasmid or empty vector, then stimulated with LPS (100 ng/ml) for the indicated times. Immunoblot analysis of total and phosphorylated p65, IKK α/β (I) (representative image), and analysis of grey intensity was shown (J) ($n \geq 3$). pMAC, peritoneal macrophage; Neu, neutrophil; EV, empty vector. Data are representative of three independent experiments and are plotted as the mean $\pm$ SD. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ vs. corresponding control.

Vangl2 inhibits LPS-induced NF- κ B activation by interacting with p65

To clarify the regulatory mechanism of Vangl2, we transfected Chinese hamster ovary (CHO) or 293T cells with NF- κ B luciferase reporter vector, with or without the Vangl2 plasmid, then treated the cells with LPS, IL-1 β , or TNF- α . We found that Vangl2 markedly inhibited NF- κ B activation induced by LPS, IL-1 β , or TNF- α in a dose-dependent manner (**Fig. 3, A-C**). Next, we sought to determine potential signaling molecules that mediated the NF- κ B-luc reporter. NF- κ B-luc activity was strongly activated by overexpression of MyD88, IRAK1, TRAF6, IKK α , IKK β or p65, but all of these activities were inhibited when Vangl2 was co-transfected at increasing concentrations (**Fig. 3 D** and Fig. S3, A-F), suggesting that Vangl2 may block NF- κ B activation at the very downstream signaling level of p65.

To test this prediction, we transfected 293T cells with HA-tagged Vangl2 together with Flag-tagged IKK α , IKK β , p65, TRAF6, IRAK1 or MyD88. Co-immunoprecipitation (co-IP) assay revealed that Vangl2 interacted strongly with IKK α , IKK β , p65 and MyD88 (**Fig. 3 E** and Fig. S3 G, H). In addition, endogenous co-IP immunoblot analyses showed that Vangl2 was strongly associated with p65 upon LPS stimulation in BMDMs, but not with IKK α , IKK β or MyD88 (**Fig. 3 F** and Fig. S3 I). Moreover, ZDOCK server predicted that Vangl2 may potentially interact with p65 by a hydrogen bond (Fig. S3 J). We further investigated the co-localization of Vangl2 and p65 by confocal microscopy and found a weak-co-localization of Vangl2 with p65 in unstimulated cells and the co-localization between Vangl2 and p65 was notably enhanced upon LPS stimulation (**Fig. 3 G** and Fig. S3 K). To determine how Vangl2 interacts with cytoplasmic p65, we isolated cytoplasm and membrane in LPS treated THP-1 cells. We found that Vangl2 interacted with p65 mainly in the cytoplasm, although most of Vangl2 located on the membrane (Fig. S3 L). Together, these data suggest that Vangl2 may interact with p65 in the cytoplasm to inhibit NF- κ B signaling.

Vangl2 comprises an N-terminal cytoplasmic tail (NT), a transmembrane (TM) domain, a Prickle-binding domain (P_kBD), and a C-terminal cytoplasmic tail (CT) (Nagaoka, Furuse, Ohtsuka, Tsuchida, & Kishi, 2019). To map the essential domains of Vangl2 that mediate its association with p65, we generated several deletion constructs of Vangl2. We found Vangl2 FL (full-length), T2 (Δ NT+TM), T3 (Δ CT), and T4 (Δ NT) interacted with the full-length p65, while Vangl2 T1 (Δ P_kBD+CT) abrogated their association (**Fig. 3 H** and Fig. S3 M), indicating that the P_kBD domain is important for the Vangl2-p65 interaction. Additionally, we constructed HA-tagged Vangl2 P_kBD plasmid and co-IP assay revealed that Vangl2 FL and P_kBD interacted with p65 (**Fig. 3 I**). Moreover, we observed that deletion of the P_kBD domain (T1), but not other domains, of Vangl2 abolished the Vangl2-mediated inhibition of NF- κ B-luc activity, and transfection of Vangl2 P_kBD domain achieved similar inhibition on p65 induced NF- κ B activation as Vangl2 FL did (**Fig. 3 J**). Collectively, these data suggest that Vangl2 suppresses NF- κ B signaling by targeting p65 through Vangl2 P_kBD domain interaction.

Vangl2 promotes the autophagic degradation of p65

Next we sought to study the physiological role of Vangl2-p65 interaction in the regulation of NF- κ B signaling by transfecting 293T cells with Flag-tagged p65, together with increasing doses of Vangl2, and found that Vangl2 dramatically decreased the protein level of p65 in a dose-dependent manner (**Fig. 4 A**). To exclude the possibility that the downregulation of p65 protein by Vangl2 was caused by the inhibition of p65 transcription, qPCR results suggested that the abundance of p65 mRNA did not change in cells with increased expression of Vangl2 (**Fig. 4 B**). Moreover, Vangl2 facilitated the degradation of the p65 phosphorylation mutants (Fig S4A), and Vangl2 phosphorylation mutants were observed to degrade p65 (Fig S4B), indicating that Vangl2 mediated p65 degradation is independent of its phosphorylation status. Since activated p65 translocates to the nucleus, we next assessed whether Vangl2 regulates the degradation of p65 in the cytoplasm or nucleus, and found that the Vangl2 interacted with p65 mainly in the cytoplasm (Fig. S4 C) and

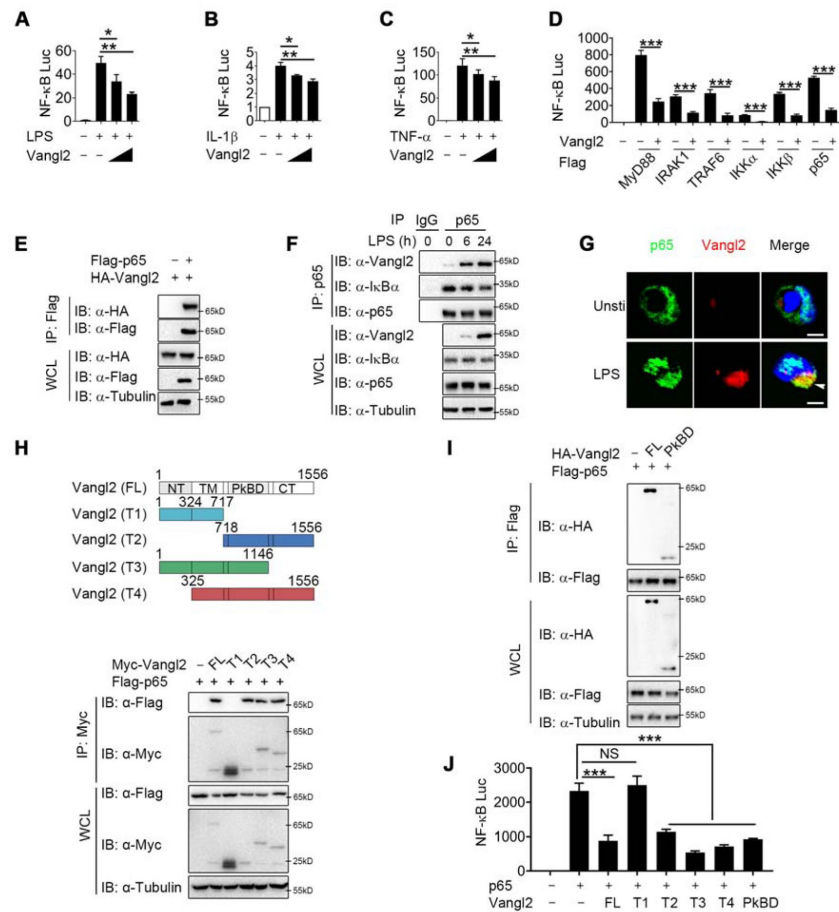


Figure 3.

Vangl2 inhibits NF-κB signaling by interacting with p65.

(A-C) Cho (A) or HEK293T cells (B and C) were co-transfected with a NF-κB and TK- Renilla reporter along with increasing amounts of Vangl2 for 18 h, then treated the cells with or without LPS (A, 250 ng/ml), IL-1β (B, 40 ng/ml), or TNF-α (C, 20 ng/ml) for 6 h. NF-κB promoter driven luciferase activity was measured and normalized to the Renilla luciferase activity. (D) Luciferase activity in HEK293T transfected with plasmids encoding an NF-κB luciferase reporter and TK-Renilla reporter, together with a vector encoding MyD88, IRAK1, TRAF6, IKKα, IKKβ, or p65, along with or without Vangl2 plasmid, was measured at 24 h after transfection and normalized to the Renilla luciferase activity. (E) HEK293T cells were transfected with plasmids encoding HA-tagged Vangl2 and Flag-tagged p65, followed by immunoprecipitation with anti-Flag beads and immunoblot analysis with anti-HA. Throughout was the immunoblot analysis of whole-cell lysates without immunoprecipitation. (F) BMDMs were stimulated with LPS (100 ng/ml) for the indicated times. The cell lysates were subjected to immunoprecipitation with an anti-p65 antibody or control IgG, followed by immunoblotting with indicated antibodies. (G) The WT and Vangl2-deficient peritoneal macrophages were treated with LPS (1000 ng/ml) for 4 h, and co-localization of p65 and Vangl2 was detected by immunofluorescence (p65, green; Vangl2, red; DAPI, blue; Scale bar, 50 μm). (H) A structural diagram of Vangl2 as well as schematic representation of Myc-tagged truncation mutants of Vangl2 (top). HEK293T cells were transfected with Flag-tagged p65 and empty vector, Myc-tagged Vangl2 (FL) or Vangl2 truncation mutants. The cell lysates were subjected to immunoprecipitation with anti-Flag beads and immunoblotted with the indicated antibodies (bottom). (I) HEK293T cells were transfected with Flag-tagged p65 and HA-tagged Vangl2 FL or PkBD truncation. The cell lysates were subjected to immunoprecipitation with anti-Flag beads and immunoblotted with the indicated antibodies. (J) Luciferase activity in HEK293T cells transfected with an NF-κB luciferase reporter, together with a vector encoding p65, along with the empty vector or with vectors encoding Vangl2 or its truncation mutants. The results are presented relative to Renilla luciferase activity.

IP, immunoprecipitation; WCL, whole-cell lysate. Data are representative of three independent experiments and are plotted as the mean ±SD. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ vs. corresponding control. NS, not significant.

mediated the degradation of p65 in the cytoplasmic fraction (**Fig. 4 C**), which is consistent with the result that Vangl2 and p65 co-localized in the cytoplasm (**Fig. 3 G**). To further assess whether Vangl2 regulates the degradation of endogenous p65, we found that Vangl2-deficient BMDMs stabilized the expression of endogenous p65 after LPS treatment (**Fig. 4 D**). After 12 h of LPS stimulation, the cycloheximide (CHX)-chase assay result showed that the degradation rate of p65 in *Vangl2^{ΔM}* BMDMs was slower than that of WT cells (Fig. S4 D). Together, these data suggest that Vangl2 promotes p65 protein degradation.

To investigate whether Vangl2 degrades p65 through an autolysosome or proteasome pathway, 293T cells were transfected with p65, together with or without the Vangl2 plasmids, and treated with different pharmacological inhibitors. We found the degradation of p65 induced by Vangl2 was blocked by autolysosome inhibitor chloroquine (CQ) and bafilomycin A1 (Baf A1), or autophagy inhibitor 3-methyladenine (3-MA) (**Fig. 4 E**), but not by the proteasome inhibitor MG132 or caspase-1 inhibitor Z-VAD and VX-765 (**Fig. 4 E** and Fig. S4 E). Furthermore, Vangl2 significantly increased the degradation of p65 during rapamycin-triggered autophagy (**Fig. 4 F**), which suggested that Vangl2 promoted autophagic degradation of p65.

To further demonstrate Vangl2 mediated p65 degradation through autophagy, we transfected WT, ATG5, and Beclin1 knockout (KO) 293T cells with Vangl2, and found that the degradation of p65 triggered by Vangl2 was almost abrogated in ATG5 and Beclin1 KO cells (**Fig. 4 G**). The p65 turnover rates were markedly reduced in ATG5 and Beclin1 KO cells post CHX treatment (Fig. S4, F and G), suggesting that the impaired autophagy prevented p65 degradation. Consistently, NF-κB activation induced by p65 was rescued in ATG5 and Beclin1 KO cells expressing Vangl2, compared to WT 293T cells (**Fig. 4, H** and **I**). In addition, the colocalization of p65 and LC3B was enhanced following LPS stimulation (Fig. S4, H). Together, our data suggest that Vangl2 specifically promotes p65 degradation through autophagy.

Vangl2 promotes the recognition of p65 by cargo receptor NDP52

Accumulating evidence showed that cargo receptors play crucial roles in selective autophagic degradation by delivering substrates (X. He et al., 2019; Kirkin & Rogov, 2019; Wu et al., 2021). Since there is no research suggests that Vangl2 is a cargo protein, we hypothesized that Vangl2 might bridge p65 to the cargo receptors for autophagic degradation. To identify the potential cargo receptor responsible for Vangl2-mediated autophagic degradation of p65, we co-transfected 293T cells with Vangl2 and various cargo receptors, followed by Co-IP assay. Result suggested that Vangl2 strongly interacted with the cargo receptors p62 and NDP52, and slightly associated with NBR1 and Nix (**Fig. 5 A**). However, p65 only interacted with the cargo receptors p62 and NDP52 (**Fig. 5 B**), which is consistent with a recent finding that p62 protein is a Vangl2-binding partner (Puvirajesinghe et al., 2016). We next attempted to clarify whether p62 or NDP52 is involved in Vangl2-mediated autophagic degradation of p65. Interestingly, we found that Vangl2 promoted the association between p65 and NDP52, but did not affect p65-p62 complex (**Fig. 5 C**). Consistently, we found that Vangl2 mediated degradation of p65 was rescued in NDP52 KO cells, but not in p62 KO cells (**Fig. 5 D**). Endogenous co-IP immunoblot analyses also revealed that deficiency of Vangl2 remarkably attenuated the association of endogenous p65 and NDP52 (**Fig. 5 E**). Likewise, NDP52, but not p62, enhanced the association between p65 and Vangl2 (**Fig. 5 F** and Fig. S5 A). And Vangl2 failed to inhibit the activation of NF-κB signaling in NDP52 KO cells (**Fig. 5 G**), but not in p62 KO cells (Fig. S5 B). Furthermore, compared with WT cells, CHX-chase assay results showed that the degradation rates of p65 were reduced in NDP52 KO cells (**Fig. 5 H**), but not in p62 KO cells (Fig. S5 C). Taken together, these data suggest that Vangl2 mediates the NDP52-directed selective autophagic degradation of p65.

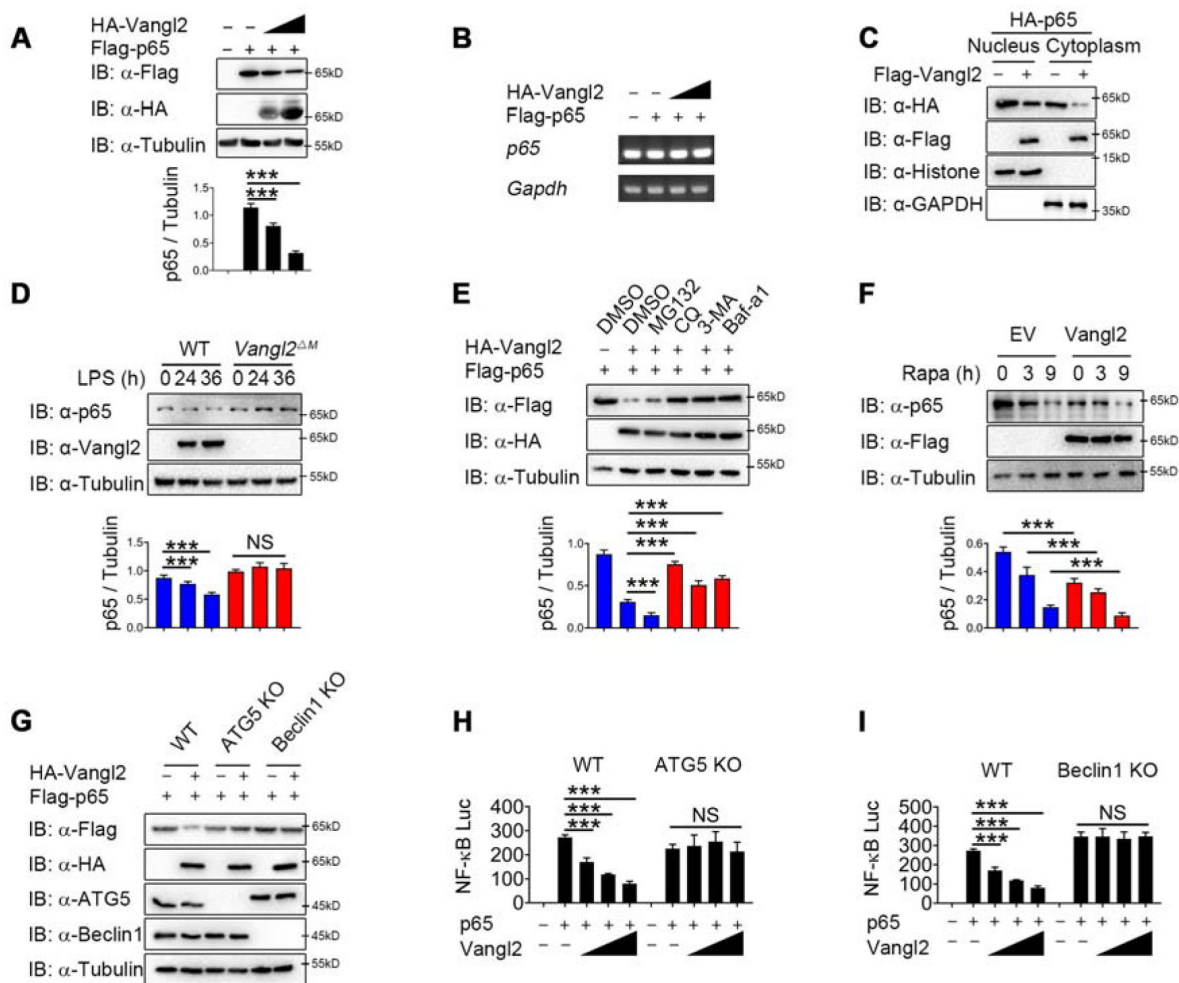


Figure 4.

Vangl2 promotes the autophagic degradation of p65.

(A) Immunoblot analysis of HEK293T cells transfected with Flag-p65 and increasing amounts of the vector encoding HA-Vangl2 (0, 250, 500, and 1000 ng).
 (B) Total RNA from HEK293T cells as in (A) was isolated and measured by semi-quantitative PCR.
 (C) HEK293T cells transfected with Flag-p65 and increasing amounts of the vector encoding HA-Vangl2, and the expressions of p65 in nuclear or cytoplasm were detected by immunoblot.
 (D) WT and Vangl2-deficient BMDMs were treated with LPS for the indicated times, and the expressions of p65 and Vangl2 were detected by immunoblot.
 (E) HEK293T cells were transfected with Flag-p65 and HA-Vangl2 plasmids, and treated with DMSO, MG132 (10 μ M), CQ (50 μ M), 3-MA (10 mM) or Baf-A1 (0.2 μ M) for 6 h. The cell lysates were analyzed by immunoblot.
 (F) HEK293T cells were transfected with empty vector (EV) or Flag-Vangl2 plasmid, and treated with rapamycin for the indicated times. The cell lysates were analyzed by immunoblot with indicated antibodies.
 (G) WT, ATG5 KO and Beclin1 KO HEK293T cells were transfected with Flag-p65, together with or without HA-Vangl2 plasmids, and then the cell lysates were analyzed by immunoblot with indicated antibodies.
 (H and I) Luciferase activity in WT, ATG5 KO (G) and Beclin1 KO (H) HEK293T cells transfected with plasmids encoding an NF- κ B luciferase reporter and TK-Renilla reporter, together with p65 plasmid along with increasing amounts of Vangl2, was measured at 24 h after transfection and normalized to the Renilla luciferase activity. CHX, cycloheximide; 3-MA, 3-methyladenine; CQ, chloroquine; Baf A1, bafilomycin A1. Data are representative of three independent experiments and are plotted as the mean $\pm$ SD. * p <0.05, ** p <0.01, *** p <0.001 vs. corresponding control. NS, not significant.

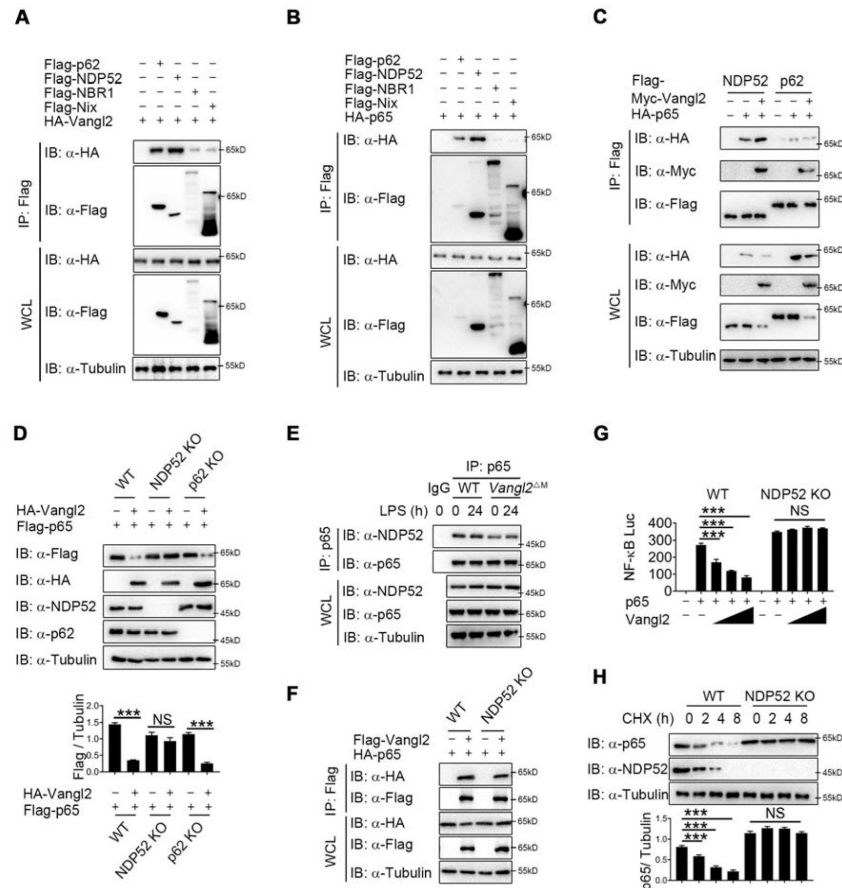


Figure 5.

Vangl2 enhances the recognition of p65 by cargo receptor NDP52.

(A) HEK293T cells transfected with a vector expressing HA-Vangl2 along with the empty vector or vector encoding Flag-p62/NDP52/NBR1/Nix. The cell lysates were subjected to immunoprecipitation with anti-Flag beads and immunoblotted with the indicated antibodies.

(B) HEK293T cells transfected with a vector expressing HA-p65 along with the empty vector or vector encoding Flag-p62/NDP52/NBR1/Nix. The cell lysates were subjected to immuno-precipitation with anti-Flag beads and immunoblotted with the indicated antibodies.

(C) HEK293T cells were transfected with HA-p65 together with Flag-NDP52 or Flag-p62, as well as with empty vector or Myc-Vangl2. The cell lysates were subjected to immunoprecipitation with anti-Flag beads and immunoblotted with the indicated antibodies.

(D) WT, NDP52 KO and p62 KO HEK293T cells were transfected with a vector expressing HA-p65 along with the empty vector or vector encoding Flag-Vangl2. The cell lysates were immunoblotted with the indicated antibodies.

(E) WT and Vangl2-deficient BMDMs were stimulated with LPS (100 ng/ml) for the indicated times. The cell lysates were subjected to immunoprecipitation with an anti-p65 antibody or control IgG and immunoblotted with the indicated antibodies.

(F) WT and NDP52 KO HEK293T cells were transfected with a vector expressing HA-p65 along with the empty vector or vector encoding Flag-Vangl2. The cell lysates were subjected to immunoprecipitation with anti-Flag beads and immunoblotted with the indicated antibodies.

(G) Luciferase activity in WT and NDP52 KO HEK293T cells transfected with plasmids encoding NF-κB luciferase reporter and TK-Renilla reporter, together with p65 plasmid along with increasing amounts of Vangl2 plasmid, was measured at 24 h after transfection.

(H) WT and NDP52 KO HEK293T were treated with CHX for the indicated times. The cell lysates were immunoblotted with the indicated antibodies.

Data are representative of three independent experiments and are plotted as the mean $\pm$ SD. * p <0.05, ** p <0.01, *** p <0.001 vs. corresponding control. NS, not significant.

Vangl2 increases the K63-linked poly-ubiquitination of p65

It has been well documented that ubiquitin chains attached to the substrates and served as a signal for the recognition by cargo receptors (Otten et al., 2021 [↗](#); Shaid, Brandts, Serve, & Dikic, 2013 [↗](#); Yin, Popelka, Lei, Yang, & Klionsky, 2020 [↗](#)). The ubiquitin-associated (UBA) domain of NDP52 mostly recognized ubiquitinated substrates for degradation through autophagy (Johansen & Lamark, 2014 [↗](#); Yamano & Youle, 2020 [↗](#)). We hypothesized that Vangl2 may affect the ubiquitin chains on p65 for subsequent NDP52-dependent degradation. To demonstrate this, we performed endogenous co-IP assay and found that the poly-ubiquitination of p65 was impaired in BMDMs from *Vangl2*^{ΔM} mice after LPS stimulation, compared to WT mice (Fig. 6 A [↗](#)). Moreover, Vangl2 specifically increased the total and K63-linked (K63-only ubiquitin mutant) poly-ubiquitination of p65 (Fig. 6 B [↗](#)). In contrast, Vangl2 did not affect other ubiquitin linkages (K11, K27, K33, or K48-only ubiquitin mutant) of p65 in an overexpression system (Fig. 6 C [↗](#)). To further substantiate the hypothesis that Vangl2 is involved in K63-linked poly-ubiquitinated p65, we conducted a knockdown experiment using siRNA to reduce Vangl2 expression. Our results demonstrated that the absence of Vangl2 inhibits K63-linked poly-ubiquitination of p65 (Fig. 6 D [↗](#)). Together, these results suggest that Vangl2 promotes the degradation of p65 by promoting the K63-linked ubiquitination of p65.

Vangl2 recruits PDLIM2 to ubiquitinate p65

Although Vangl2 promotes K63-linked ubiquitination and degradation of p65, Vangl2 is not an E3 ubiquitin ligase. We hypothesized that Vangl2 might function as a scaffold protein to link p65 and its E3 ubiquitin ligase for ubiquitination or to block the interaction of deubiquitinase (DUB) with p65. To identify the potential E3 ubiquitinase or DUB responsible for Vangl2 mediated ubiquitination of p65, we further analyzed our RNA-seq data to identify E3 ubiquitin ligase or DUB involved in ubiquitination of p65 by comparing LPS-stimulated BMDMs from *Vangl2*^{ΔM} and WT mice. The result of differentially expression identified 88 down-regulated genes related to E3 ubiquitin ligase and 56 up-regulated genes related to DUB in response to Vangl2 deficiency in BMDMs after LPS stimulation (Fig. S6 A), which included PDZ-LIM domain-containing protein 2 (PDLIM2), Trim21 and DUB ubiquitin-specific peptidase 7 (USP7). Recent research showed that ubiquitin E3 ligases PDLIM2 and Trim21 ubiquitinated p65 via K63 linkage and enhanced the interaction of p65 with IKK (Healy & O'Connor, 2009 [↗](#); Jodo, Shibazaki, Onuma, Kaisho, & Tanaka, 2020 [↗](#); Yang et al., 2021 [↗](#)), while DUB USP7 promoted NF-κB-mediated transcription (Mitxitorena et al., 2020 [↗](#)). Here, we found that the mRNA level of PDLIM2 and USP7 decreased in Vangl2-deficient BMDMs, compared with WT BMDMs (Fig. S6, B-D). To investigate which E3 ubiquitin ligase or DUB is recruited by Vangl2, luciferase assay in Vangl2-expressing 293T cells transfected with scramble, *Pdlim2*, *Usp7*, or *Trim21* siRNA (Fig. S6, E-G) suggested Vangl2-mediated inhibition of NF-κB activation and degradation of p65 were blocked by knocking down PDLIM2, but not USP7 or Trim21 (Fig. 7 A [↗](#)). Meanwhile, *Pdlim2* knockdown in BMDMs also resulted in higher expression of *Il6* and *Il1b* in response to LPS stimulation (Fig. 7, B [↗](#) and C [↗](#)), indicating that PDLIM2 plays a key role in promoting Vangl2-mediated p65 degradation.

We next investigated whether Vangl2 promoted the association between p65 and PDLIM2 by co-IP experiments. Our results indicated that Vangl2 interacted with PDLIM2 (Fig. 7 D [↗](#)) and promoted the association between p65 and PDLIM2 (Fig. 7 E [↗](#)). Notably, PDLIM2 accelerated the degradation of p65 in the presence of Vangl2 (Fig. 7 F [↗](#)). Conversely, PDLIM2 deficiency markedly impaired Vangl2-mediated K63-linked ubiquitination of p65 (Fig. 7 G [↗](#)) and knockdown of Vangl2 inhibited K63-linked poly-ubiquitination of p65 mediated by PDLIM2 (Fig. 7 H [↗](#)). Taken together, these data suggest that Vangl2 functions as a potential adaptor to recruit E3 ubiquitin ligase PDLIM2 to p65 and promotes the K63-linked ubiquitination of p65 for its subsequent autophagic degradation.

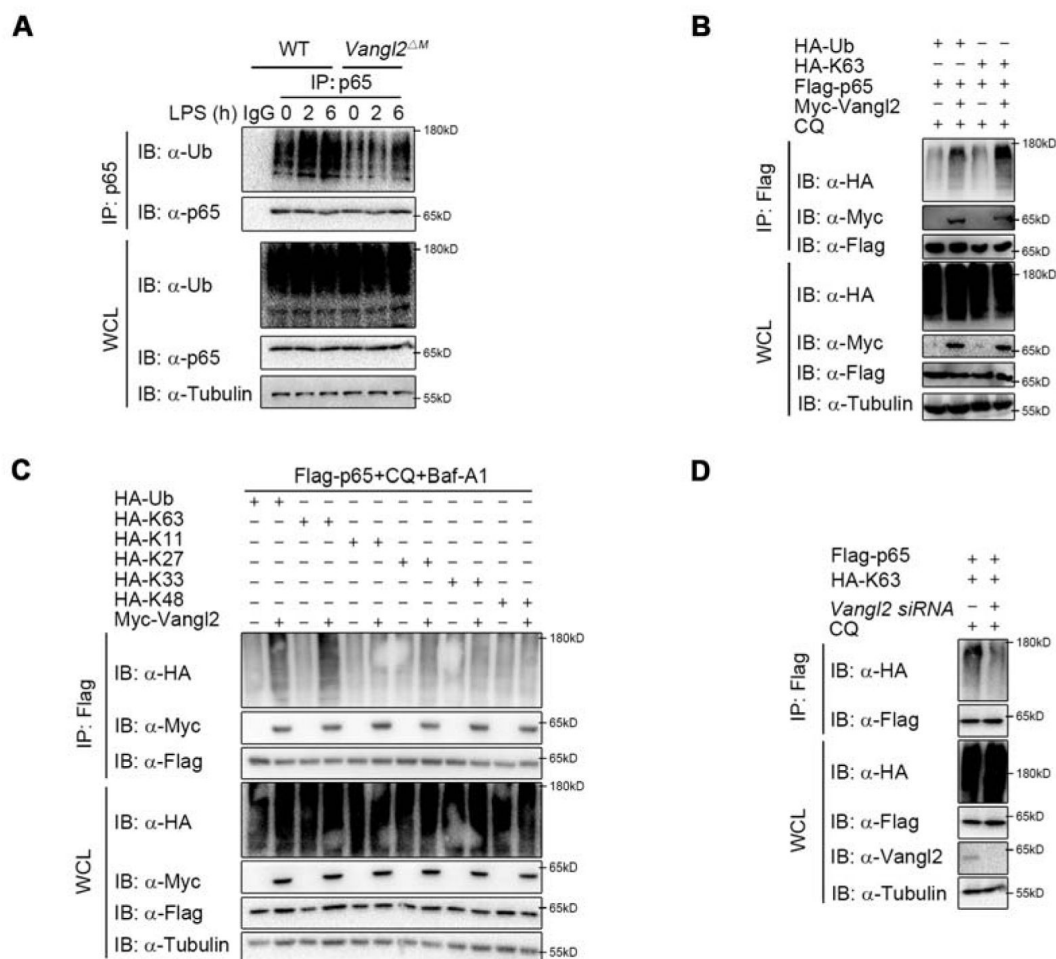


Figure 6.

Vangl2 increases the K63-linked ubiquitination of p65.

(A) WT and *Vangl2*-deficient BMDMs were stimulated with LPS (100 ng/ml) for the indicated times. The cell lysates were subjected to immunoprecipitation with an anti-p65 antibody or control IgG and immunoblotted with the indicated antibodies.

(B) HEK293T cells were transfected with Flag-p65, Myc-Vangl2, HA-Ub or HA-K63 plasmids with the indicated combinations for 24 h and then treated with CQ for 8 h. The cell lysates were subjected to immunoprecipitation with anti-Flag beads and immunoblotted with the indicated antibodies.

(C) HEK293T cells were transfected with Flag-p65, Myc-Vangl2 and HA- Ub/K63/K11/K27/K33/K48 plasmids with the indicated combinations for 24 h and then treated with CQ and Baf-A1 for 8 h. The cell lysates were subjected to immunoprecipitation with anti-Flag beads and immunoblotted with the indicated antibodies.

(D) HEK293T cells were transfected with a vector expressing Flag-p65 and HA-K63 along with *Scramble* or *Vangl2* siRNA. The cell lysates were subjected to immunoprecipitation with anti-Flag beads and immunoblotted with the indicated antibodies. Data are representative of three independent experiments.

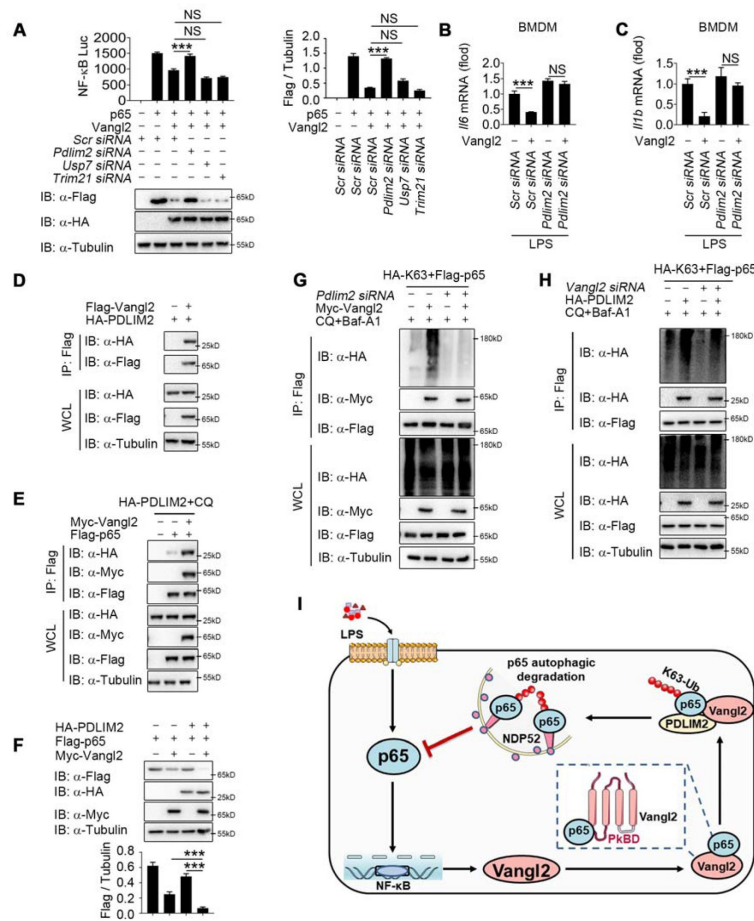


Figure 7.

Vangl2 recruits PDLIM2 to ubiquitinate p65.

(A) HEK293T cells were transfected with the indicated siRNA, NF-κB reporter plasmids together with HA-Vangl2, Flag-p65 or the control vector as indicated for 24 h, and then subjected to luciferase assay and immunoblotting analysis.

(B and C) BMDMs were transfected with *Pdlim2* or *Scramble* siRNA along with the empty vector or vector encoding Flag-Vangl2, stimulated with LPS (100 ng/ml) for 6 h, then analyzed by qPCR for *Il6* **(B)** and *Il1b* **(C)** expression.

(D) HEK293T cells transfected with HA-PDLIM2 along with the empty vector or vector encoding Flag-Vangl2. The cell lysates were subjected to immunoprecipitation with anti-Flag beads and immunoblotted with the indicated antibodies.

(E) HEK293T cells were transfected with Flag-p65, HA-PDLIM2 and Myc-Vangl2 plasmids with the indicated combinations for 24 h and then treated with CQ and Baf-A1 for 8 h. The cell lysates were subjected to immunoprecipitation with anti-Flag beads and immunoblotted with the indicated antibodies.

(F) HEK293T cells were transfected with Flag-p65, HA-PDLIM2 and Myc-Vangl2 plasmids with the indicated combinations for 24 h. The cell lysates were immunoblotted with the indicated antibodies.

(G) HEK293T cells were transfected with Flag-p65, HA-K63 and Myc-Vangl2 plasmids, the expression of E3 ubiquitin ligase was interfered with *Pdlim2* siRNA and then treated with CQ and Baf-A1 for 8 h. The cell lysates were subjected to immunoprecipitation with anti-Flag beads and immunoblotted with the indicated antibodies.

(H) HEK293T cells were transfected with *Pdlim2* or *Scramble* siRNA, along with or without HA-PDLIM2, then treated with CQ and Baf-A1 for 8 h. The cell lysates were subjected to immunoprecipitation with anti-Flag beads and immunoblotted with the indicated antibodies.

(I) A schematic model to illustrate how Vangl2-PDLIM2-NDP52-p65 axis negatively regulates NF-κB activation. During LPS stimulation, Vangl2 expression is up-regulated, thus constituting a negative feedback loop to regulate NF-κB activation. In detail, Vangl2 functions as an adaptor protein to recruit an E3 ubiquitin ligase PDLIM2 to increase K63-linked ubiquitination of p65 and promotes NDP52-mediated p65 degradation through selective autophagy, resulting in ameliorating sepsis and suppressing production of pro-inflammatory cytokines.

Data are representative of three independent experiments and are plotted as the mean ±SD. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ vs. corresponding control. NS, not significant.

Discussion

As a core PCP component, Vangl2 is widely known for its function in organ development, such as brain, tooth, tongue and kidney (Bailly et al., 2018 [DOI](#); Hatakeyama et al., 2014 [DOI](#)), trafficking from the endoplasmic reticulum (ER) to the cell surface, and subsequently shuttles between the endocytic vesicles and cell surface (D. Feng et al., 2021 [DOI](#)). The function of Vangl2 mostly depends on its cellular localization (Hatakeyama et al., 2014 [DOI](#)) and is reported mainly through PCP/WNT signaling pathway (Jessen & Jessen, 2019 [DOI](#)). Activated Vangl2 exhibits extremely long cytoplasmic and intercellular branches and delivers Wnt to multiple cells to enhance Wnt/ β -Catenin signaling (Brunt et al., 2021 [DOI](#)). During myocardial hypertrophy, Vangl2 aggravates myocardial hypertrophy by regulating Wnt/JNK signaling (Brunt et al., 2021 [DOI](#); Jessen & Jessen, 2017 [DOI](#), 2019 [DOI](#)) and the expansion of cardiomyocyte surface area (Brunt et al., 2021 [DOI](#); Jessen & Jessen, 2017 [DOI](#), 2019 [DOI](#)). However, our previous study showed that the lysosome-suppressing function of Vangl2 in osteoblast differentiation is not dependent on conventional PCP pathway (Gong et al., 2021 [DOI](#)), and we found Vangl2 did not affect the JNK pathway after LPS treatment (data not shown), suggesting that Vangl2 has additional functions in pathways besides PCP. In this study, we provided direct evidence that the expression of Vangl2 was increased during sepsis and upregulated significantly in immune organs (lymph nodes and spleen) upon LPS stimulation, which is consistent with our previous finding that Vangl2 regulated the downstream signaling of TLR or IL-1R. In addition, the present study showed that Vangl2 prevented the progression of sepsis and the accumulation of inflammatory cytokines through suppressing NF- κ B pathway: Vangl2 inhibited LPS-induced NF- κ B activation by delivering p65 to autophagosome for degradation. To the best of our knowledge, this is the first study proves that Vangl2 regulates NF- κ B signaling and inflammatory responses, and could serve as a potential target for therapeutic purposes in diseases associated with NF- κ B signaling.

As is well-known, Vangl2 phosphorylation is essential for its functions in the mammalian PCP pathway in multiple tissues. However, our findings indicate that Vangl2 phosphorylation mutants also promote p65 degradation, suggesting that Vangl2 regulates the NF- κ B pathway independently of its conventional PCP pathway. The PCP pathways frequently intersect with developmentally important pathways. Previous research has demonstrated that Vangl2 deficiency can affect developmental processes. It would be importance to investigate whether Vangl2-dependent NF- κ B is influenced by developmental pathways. Distinct from previous studies, our findings revealed no significant differences in spleen and lymph node size between wild-type (WT) and *Vangl2* ^{Δ M} mice. However, myeloid-specific loss of Vangl2 resulted in increased numbers of monocytes, macrophages, and neutrophils in the spleen and bone marrow. This suggests that the ablation of Vangl2 in myeloid cells does not affect the development of immune system development, but rather affects the differentiation of specific cell types. Moreover, we found that Vangl2 is induced and functions under LPS stimulation. Collectively, these findings suggest that Vangl2-dependent NF- κ B functions do not affect the development of the immune system, but rather influence the progression of inflammatory-related diseases.

The proteasome, lysosome and autolysosome pathways are the major systems that are utilized by eukaryotic cells to maintain the protein abundance and immune homeostasis (Deretic, 2021 [DOI](#)). Previous studies have shown that cellular levels of Vangl2 (D. Feng et al., 2021 [DOI](#)) and its binding partner Prickle2 (Nagaoka et al., 2019 [DOI](#)) are maintained by the proteasomal pathway. Our recent study showed that Vangl2 mediated osteogenic differentiation by limiting chaperone-mediated autophagy (CMA) in mesenchymal stem cells, suggesting a close relationship between Vangl2 and autophagy (Gong et al., 2021 [DOI](#)). Indeed, our study suggests Vangl2 promotes p65 degradation through autophagy pathway, but not proteasomal pathway. Importantly, we demonstrated that the degradation of p65 mediated by Vangl2-NDP52 complex is regulated by autophagy induction

through rapamycin treatment and autophagy blockade by ATG5 KO or Beclin1 KO. Considering our findings and other reports, Vangl2 may play multi-functional roles in regulating different types of autophagy (i.e. CMA or selective autophagy).

Selective autophagy requires that cargo receptors recognize the labeling of cargoes with degradation signals and subsequently engaged with the LC3 localized in the autophagosome membrane (Shaid et al., 2013 [\[1\]](#)). Common cargo receptors mainly include p62/SQSTM1, CALCOCO2/NDP52, OPTN, NBR1 and TOLLIP (Johansen & Lamark, 2014 [\[2\]](#); Yamano & Youle, 2020 [\[3\]](#)). A study in HEK293T cells showed that LRRC25 promotes the autophagic degradation of p65 through enhancing the interaction between p65 and p62 (Y. Feng et al., 2017 [\[4\]](#)). In addition, a recent study showed that Vangl2 interacts with p62, subsequently promoting breast tumors by activating JNK signaling (Puvirajesinghe et al., 2016 [\[5\]](#)). Thus, we hypothesized that Vangl2 may promote the autophagic degradation of p65 by recruiting the cargo receptor p62. However, our data suggested that Vangl2 markedly increased the p65-NDP52 interaction but not p65-p62. Strikingly, Vangl2 mediated autophagic degradation of p65 was abolished in NDP52 KO cells, but not in p62 KO cells. Thus, our findings identify that NDP52 is the new cargo receptor responsible for Vangl2-mediated selective autophagic degradation of p65.

Accumulating evidence has shown that cargo receptors mainly recognize ubiquitination modifications on the substrates and then promote degradation in auto- lysosome (Shaid et al., 2013 [\[1\]](#)). As expected, we observed Vangl2 promoted the ubiquitination of p65, thus enhancing the association between cargo protein and p65. K48- and K63-linked ubiquitination were mostly reported modifications on p65 (Kauppinen, Suuronen, Ojala, Kaarniranta, & Salminen, 2013 [\[6\]](#); Korbecki, Bobinski, & Dutka, 2019 [\[7\]](#)). For example, E3 ubiquitin ligase RNF182 inhibited TLR triggered cytokine production by promoting K48-linked poly-ubiquitination of p65 (Cao, Sun, Chang, Sun, & Yang, 2019 [\[8\]](#)). Trim21 promoted K63-linked poly-ubiquitination of p65, but did not affect the stability of p65 (Yang et al., 2021 [\[9\]](#)). PDLIM7 cooperated with PDLIM2 to inhibit inflammation by promoting K63-linked poly-ubiquitination of p65 (Jodo et al., 2020 [\[10\]](#)). In this study, we revealed that Vangl2 specifically promoted K63-linked poly-ubiquitination of p65, but not other ubiquitin linkages by recruiting a previously unrecognized E3 ligase PDLIM2, which further explore the molecular mechanism by which regulates the stability of p65.—Although many studies focus on the regulation of p65, the location of p65 degradation has not been clarified. A study showed PDLIM2 degrades p65 through the proteasomal pathway in the nucleus (Jodo et al., 2020 [\[10\]](#)), while we found Vangl2 mediated autophagic degradation of p65 mainly happened in cytoplasm, but not in nucleus. Combined with previous studies, our study indicates that p65 undergoes different degradation pathway in distinct location of cells.

Autophagy is a fundamental biological process contributing to multiple life processes. Emerging evidence has suggested that crosstalk between autophagy and innate immune signaling plays critical roles in diseases with inflammatory components, including infections, cancer, autoimmunity, and metabolic disorders (Pradel et al., 2020 [\[11\]](#); Wu et al., 2021 [\[12\]](#)). Recent study showed that the interplay between autophagy and type I IFN or NF- κ B signaling drives or suppresses inflammatory responses during SARS-CoV-2 infection (Hui et al., 2021 [\[13\]](#)). Moreover, autophagy related key genes, such as Atg5, Atg9, and ULK1, also play critical roles in inflammatory diseases (Y. He et al., 2018 [\[14\]](#); Peng et al., 2019 [\[15\]](#)). Here, our discovery of Vangl2-PDLIM2-NDP52 complex in the regulation of p65-mediated NF- κ B signaling could be a therapeutic target for the development of immunotherapy against infection and inflammation.

Based on our findings, we propose a working model that Vangl2 negatively regulates NF- κ B signaling (**Fig. 7 I** [\[16\]](#)). Vangl2 functions as an adaptor to recruit ubiquitin ligase PDLIM2 and increase K63-linked ubiquitination on p65, which promotes the recognition of p65 by cargo receptor NDP52 and the autophagic degradation of p65, resulting in suppressing the production of pro-inflammatory cytokines and ameliorating sepsis. Our findings provide a potential target for the treatment of inflammatory diseases.

Materials and Methods

Animal and sepsis model

All animal experiments were approved by the Southern Medical University Animal Care and Use Committee (SMUL20201010). The *Vangl2*^{flox/flox} mice and lysozyme-Cre (*Lyz2*-Cre) mice were obtained from the Jackson Laboratory (Bar Harbor, ME, USA; Jax no. 025174 and 004781, respectively). *Vangl2*^{flox/flox} mice were hybridized with *Lyz2*-Cre mice to obtain *Vangl2*^{flox/flox}*Lyz2*-Cre (*Vangl2*^{ΔM}) mice with *Vangl2* specific deficiency in myeloid cells. Co-housed littermate controls with normal *Vangl2* expression were used as wild type (WT). For sepsis model, 10-weeks-old mice were intraperitoneally (i.p.) injected with LPS (25–30 mg/kg), and survival rate of mice was continuously observed. For other detection, mice with sepsis were sacrificed and samples were collected at indicated time points.

Isolation of immune cells

For bone marrow (BM)-derived macrophages (BMDMs), BM cells were collected from the femur and tibia, and then maintained in 20% L929 conditioned media with 1% penicillin-streptomycin and macrophage-colony stimulating factor (M-CSF) (St Louis, MO, USA) for 6 days, as previously described (Tan et al., 2019 [DOI](#)). Mouse peritoneal macrophages were acquired from ascites of indicated mice, which were administrated intraperitoneally (i.p.) with 4% (v/v) thioglycollate (BD) for three consecutive days before sacrifice. As for primary neutrophils, mice were administrated (i.p.) with 4% (v/v) thioglycollate (BD) for 4 h before sacrifice. Peritoneal cavities were flushed with PBS to obtain peritoneal macrophages or neutrophils. Cells were cultured in complete DMEM (Corning) supplemented with 100 U/ml penicillin (Sigma), 100 μg/ml streptomycin (Sigma) and 10% FBS (HyClone) for 6 h and then washed twice with PBS to remove adherent cells. For human PBMC isolation, 4 ml of whole blood was harvested from each sepsis patient and healthy control. The white membrane layer was collected after density gradient centrifugation at 400 g for 25 min and the PBMCs were filtered and prepared for the subsequent experiments after the removal of the red blood cells. For mouse CD11b⁺ splenocytes isolation, spleens were digested in 2% FBS-DMEM with 200 U/ml DNase I (Sigma, USA) and 1 mg/ml collagenase type II (Sigma, USA) at 37 °C for 30 min. Then tissues digested were filtered through a 70 μm cell strainer and red cells were removed by ACK. The rest splenocytes were labeled with anti-mouse CD11b biotin antibodies (Biolegend). The mixture above was then incubated with streptavidin-paramagnetic particles (BD Biosciences) at 4 °C for 30 min. Purification of CD11b⁺ splenocytes was performed by DynaMag™ (Thermo Fisher Science). Isolated cells achieved a purity of ≥95% measured by FACS.

Cell isolation and culture

HEK293T, A549, CHO and Raw 264.7 cells were obtained from the American Type Culture Collection (ATCC). Above cell lines were maintained in complete DMEM, supplemented with 10% FBS, 100 U/ml penicillin and streptomycin. All cells were cultured at 37 °C with 5% CO₂.

Cell treatment

To test cytokines expression and signaling pathway activation, BMDMs, neutrophils and A549 cells were treated with LPS (200 ng/ml) for the indicated time. For dual luciferase assay, HEK293T cells were treated with LPS (250 ng/ml), IL-1β (40 ng/ml) or TNF-α (20 ng/ml) for 6 h. For protein degradation inhibition assays in HEK293T cells, CQ (50 μM), 3MA (10 mM), or bafilomycin A1 (Baf A1) (0.2 μM) was used to inhibit autolysosome- or lysosome-mediated protein degradation. MG132 (10 μM) was used to inhibit proteasome-mediated protein degradation. Z-VAD-FMK (50 μM) was used to inhibit Caspase-mediated protein degradation.

Luciferase and reporter assays

HEK293T cells were plated in 24-well plates and transfected with pRL-TK and plasmids encoding the NF- κ B luciferase reporter, together with different plasmids following: Flag- MyD88, Flag-IRAK1, Flag-TRAF6, Flag-IKK α , Flag-IKK β , Flag-p65 and an increasing doses of the HA-Vangl2 vector (250, 500, 1000 ng) or empty vector. In addition, CHO or HEK293T cells were administrated with or without LPS (1000 ng/ml), IL-1 β (1000 ng/ml) or TNF- α (100 ng/ml) for 6-8 h after transfection with pRL-TK and plasmids encoding the NF- κ B luciferase reporter. Then, cells were collected at 24 h post-transfection and luciferase activity was analyzed by Dual-Luciferase Reporter Assay Kit (Vazyme) performed with a Luminoskan Ascent luminometer (Thermo Fisher Scientific). The activity of Firefly luciferase was normalized by that of Renilla luciferase to obtain relative luciferase activity.

Enzyme-linked immunosorbent assay (ELISA)

IL-1 β , IL-6 and TNF- α in cell supernatants and mice serum were measured using ELISA kit (#E-EL-M0037c, #E-EL-M0044c, #E-EL-M1084c, respectively; Elabscience Biotechnology) following the manufacturer's instructions. Absorbance was detected at 450 nm by the Multiscan FC (Thermo Fisher, Waltham, MA, USA).

Statistical Analysis

The data of all quantitative experiments are presented as mean $\pm$ SD of at least three independent experiments. Curve data was assessed by GraphPad Prism 8.0 (USA). And comparisons between groups for statistically significant differences were analyzed with a two-tailed Student's t test. The statistical significance was defined as $p < 0.05$.

Online supplemental material

Fig. S1 shows that the expression of Vangl2 during sepsis and LPS treatment.

Fig. S2 shows that Vangl2 defection promotes LPS-induced NF- κ B activation and production of inflammatory cytokines.

Fig. S3 shows that Vangl2 interacts with p65 to inhibit NF- κ B activation.

Fig. S4 shows that Vangl2 promotes p65 degradation by autophagic pathway.

Fig. S5 shows that Vangl2 promoted autophagic degradation of p65 is not mediated by cargo receptor p62.

Fig. S6 shows that expression of candidate E3 ubiquitin ligases in WT and Vangl2- deficient BMDMs after LPS treatment.

Table S1 shows reagents and antibodies used in this study. Table S2 shows primers sequences for quantitative RT-PCR.

Acknowledgements

We thank Dr. Jun Cui (Sun Yat-sen University) for providing ATG5, Beclin1, NDP52, and p62 KO HEK293T cells. We thank Experimental Education/Administration Center of School of Basic Medical Science of Southern Medical University for providing assistance. This work was supported by grants from National Natural Science Foundation of China (82171741, 81801579 and 82302731), Guangdong Basic and Applied Basic Research Foundation (2021A1515012140), and China Postdoctoral Science Foundation (2021M701622).

Author Contributions

J.L., J.Z., H.J. and Z.H. designed and performed the experiments. Y.Z., L.H., D.W., K.Z., P.T., and Z.G. provided assistance or technical support in some experiments. H. L., Y. X., J. Y., Q. X. and Z. S. collected samples. J. L. performed statistical analysis. J.L., J.Z., Z.H., C.P. X.B. and X.Y. performed data analysis and wrote the manuscript. X.Y. supervised the entire project.

References

- Ashrafi G., Schwarz T. L. (2013) **The pathways of mitophagy for quality control and clearance of mitochondria** *Cell Death Differ* **20**:31–42 <https://doi.org/10.1038/cdd.2012.81>
- Bailly E., Walton A., Borg J. P. (2018) **The planar cell polarity Vangl2 protein: From genetics to cellular and molecular functions** *Semin Cell Dev Biol* **81**:62–70 <https://doi.org/10.1016/j.semcdb.2017.10.030>
- Bartuzi P., Hofker M. H., van de Sluis B. (2013) **Tuning NF-kappaB activity: a touch of COMMD proteins** *Biochim Biophys Acta* **1832**:2315–2321 <https://doi.org/10.1016/j.bbadis.2013.09.014>
- Brunt L., Greicius G., Rogers S., Evans B. D., Virshup D. M., Wedgwood K. C. A., Scholpp S. (2021) **Vangl2 promotes the formation of long cytonemes to enable distant Wnt/beta-catenin signaling** *Nat Commun* **12** <https://doi.org/10.1038/s41467-021-22393-9>
- Cancer Genome Atlas Research, N., Kandoth C., Schultz N., Cherniack A. D., Akbani R., Liu Y., Levine D. A. (2013) **Integrated genomic characterization of endometrial carcinoma** *Nature* **497**:67–73 <https://doi.org/10.1038/nature12113>
- Cao Y., Sun Y., Chang H., Sun X., Yang S. (2019) **The E3 ubiquitin ligase RNF182 inhibits TLR-triggered cytokine production through promoting p65 ubiquitination and degradation** *FEBS Lett* **593**:3210–3219 <https://doi.org/10.1002/1873-3468.13583>
- Cartwright T., Perkins N. D., C. L. W. (2016) **NFKB1: a suppressor of inflammation, ageing and cancer** *FEBS J* **283**:1812–1822 <https://doi.org/10.1111/febs.13627>
- Cerami E., Gao J., Dogrusoz U., Gross B. E., Sumer S. O., Aksoy B. A., Schultz N. (2012) **The cBio cancer genomics portal: an open platform for exploring multidimensional cancer genomics data** *Cancer Discov* **2**:401–404 <https://doi.org/10.1158/2159-8290.CD-12-0095>
- Cui J., Zhu L., Xia X., Wang H. Y., Legras X., Hong J., Wang R. F. (2010) **NLRC5 negatively regulates the NF-kappaB and type I interferon signaling pathways** *Cell* **141**:483–496 <https://doi.org/10.1016/j.cell.2010.03.040>
- Denton D., Kumar S. (2019) **Autophagy-dependent cell death** *Cell Death Differ* **26**:605–616 <https://doi.org/10.1038/s41418-018-0252-y>
- Deretic V. (2021) **Autophagy in inflammation, infection, and immunometabolism** *Immunity* **54**:437–453 <https://doi.org/10.1016/j.immuni.2021.01.018>
- Feng D., Wang J., Yang W., Li J. Y., Lin X. C., Zha F. Z., Gao B. (2021) **Regulation of Wnt/PCP signaling through p97/VCP-KBTBD7-mediated Vangl ubiquitination and endoplasmic reticulum-associated degradation** *SCIENCE ADVANCES* **7** <https://doi.org/10.1126/sciadv.abg2099>
- Feng Y., Duan T., Du Y., Jin S., Wang M., Cui J., Wang R. F. (2017) **LRRC25 Functions as an Inhibitor of NF-kappaB Signaling Pathway by Promoting p65/RelA for Autophagic Degradation** *Sci Rep* **7** <https://doi.org/10.1038/s41598-017-12573-3>

- Funes S. C., Rios M., Escobar-Vera J., Kalergis A. M (2018) **Implications of macrophage polarization in autoimmunity** *Immunology* **154**:186–195 <https://doi.org/10.1111/imm.12910>
- Gao J., Aksoy B. A., Dogrusoz U., Dresdner G., Gross B., Sumer S. O., Schultz N (2013) **Integrative analysis of complex cancer genomics and clinical profiles using the cBioPortal** *Sci Signal* **6** <https://doi.org/10.1126/scisignal.2004088>
- Gatica D., Lahiri V., Klionsky D. J (2018) **Cargo recognition and degradation by selective autophagy** *Nat Cell Biol* **20**:233–242 <https://doi.org/10.1038/s41556-018-0037-z>
- Gong Y., Li Z., Zou S., Deng D., Lai P., Hu H., Zou Z (2021) **Vangl2 limits chaperone-mediated autophagy to balance osteogenic differentiation in mesenchymal stem cells** *Dev Cell* **56**:2103–2120 <https://doi.org/10.1016/j.devcel.2021.06.011>
- Hatakeyama J., Wald J. H., Printsev I., Ho H. Y., Carraway K. L (2014) **Vangl1 and Vangl2: planar cell polarity components with a developing role in cancer** *Endocr Relat Cancer* **21**:R345–356 <https://doi.org/10.1530/ERC-14-0141>
- He X., Zhu Y., Zhang Y., Geng Y., Gong J., Geng J., Zhong H (2019) **RNF34 functions in immunity and selective mitophagy by targeting MAVS for autophagic degradation** *EMBO J* **38** <https://doi.org/10.15252/embj.2018100978>
- He Y., She H., Zhang T., Xu H., Cheng L., Yepes M., Mao Z (2018) **p38 MAPK inhibits autophagy and promotes microglial inflammatory responses by phosphorylating ULK1** *J Cell Biol* **217**:315–328 <https://doi.org/10.1083/jcb.201701049>
- Healy N. C., O'Connor R (2009) **Sequestration of PDLIM2 in the cytoplasm of monocytic/macrophage cells is associated with adhesion and increased nuclear activity of NF-kappaB** *J Leukoc Biol* **85**:481–490 <https://doi.org/10.1189/jlb.0408238>
- Hui X., Zhang L., Cao L., Huang K., Zhao Y., Zhang Y., Jin M (2021) **SARS-CoV-2 promote autophagy to suppress type I interferon response** *Signal Transduct Target Ther* **6** <https://doi.org/10.1038/s41392-021-00574-8>
- Jessen T. N., Jessen J. R (2017) **VANGL2 interacts with integrin alphav to regulate matrix metalloproteinase activity and cell adhesion to the extracellular matrix** *Exp Cell Res* **361**:265–276 <https://doi.org/10.1016/j.yexcr.2017.10.026>
- Jessen T. N., Jessen J. R (2019) **VANGL2 protein stability is regulated by integrin alphav and the extracellular matrix** *Exp Cell Res* **374**:128–139 <https://doi.org/10.1016/j.yexcr.2018.11.017>
- Jodo A., Shibasaki A., Onuma A., Kaisho T., Tanaka T (2020) **PDLIM7 Synergizes With PDLIM2 and p62/Sqstm1 to Inhibit Inflammatory Signaling by Promoting Degradation of the p65 Subunit of NF-kappaB** *Front Immunol* **11** <https://doi.org/10.3389/fimmu.2020.01559>
- Johansen T., Lamark T (2014) **Selective autophagy mediated by autophagic adapter proteins** *Autophagy* **7**:279–296 <https://doi.org/10.4161/auto.7.3.14487>
- Kauppinen A., Suuronen T., Ojala J., Kaarniranta K., Salminen A (2013) **Antagonistic crosstalk between NF-kappaB and SIRT1 in the regulation of inflammation and metabolic disorders** *Cell Signal* **25**:1939–1948 <https://doi.org/10.1016/j.cellsig.2013.06.007>

- Kirkin V., Rogov V. V (2019) **A Diversity of Selective Autophagy Receptors Determines the Specificity of the Autophagy Pathway** *Mol Cell* **76**:268–285 <https://doi.org/10.1016/j.molcel.2019.09.005>
- Korbecki J., Bobinski R., Dutka M. (2019) **Self-regulation of the inflammatory response by peroxisome proliferator-activated receptors** *Inflamm Res* **68**:443–458 <https://doi.org/10.1007/s00011-019-01231-1>
- Liu T., Zhang L., Joo D., Sun S. C (2017) **NF-kappaB signaling in inflammation** *Signal Transduct Target Ther* **2** <https://doi.org/10.1038/sigtrans.2017.23>
- Locati M., Curtale G., Mantovani A (2020) **Diversity, Mechanisms, and Significance of Macrophage Plasticity** *Annu Rev Pathol* **15**:123–147 <https://doi.org/10.1146/annurev-pathmechdis-012418-012718>
- Mao X., Gluck N., Li D., Maine G. N., Li H., Zaidi I. W., Burstein E (2009) **GCN5 is a required cofactor for a ubiquitin ligase that targets NF-kappaB/RelA** *Genes Dev* **23**:849–861 <https://doi.org/10.1101/gad.1748409>
- Mitxitorena I., Somma D., Mitchell J. P., Lepisto M., Tyrchan C., Smith E. L., Carmody R. J (2020) **The deubiquitinase USP7 uses a distinct ubiquitin-like domain to deubiquitinate NF-kB subunits** *J Biol Chem* **295**:11754–11763 <https://doi.org/10.1074/jbc.RA120.014113>
- Nagaoka T., Furuse M., Ohtsuka T., Tsuchida K., Kishi M (2019) **Vangl2 interaction plays a role in the proteasomal degradation of Prickle2** *Sci Rep* **9** <https://doi.org/10.1038/s41598-019-39642-z>
- Otten E. G., Werner E., Crespillo-Casado A., Boyle K. B., Dharamdasani V., Pathe C., Randow F (2021) **Ubiquitylation of lipopolysaccharide by RNF213 during bacterial infection** *Nature* **594**:111–116 <https://doi.org/10.1038/s41586-021-03566-4>
- Papakrivopoulou E., Vasilopoulou E., Lindenmeyer M. T., Pacheco S., Brzoska H. L., Price K. L., Long D. A (2018) **Vangl2, a planar cell polarity molecule, is implicated in irreversible and reversible kidney glomerular injury** *J Pathol* **246**:485–496 <https://doi.org/10.1002/path.5158>
- Peng X., Wang Y., Li H., Fan J., Shen J., Yu X., Mao H (2019) **ATG5-mediated autophagy suppresses NF-kappaB signaling to limit epithelial inflammatory response to kidney injury** *Cell Death Dis* **10** <https://doi.org/10.1038/s41419-019-1483-7>
- Poobalasingam T., Yates L. L., Walker S. A., Pereira M., Gross N. Y., Ali A., Dean C. H (2017) **Heterozygous Vangl2(Looptail) mice reveal novel roles for the planar cell polarity pathway in adult lung homeostasis and repair** *Dis Model Mech* **10**:409–423 <https://doi.org/10.1242/dmm.028175>
- Pradel B., Robert-Hebmann V., Espert L (2020) **Regulation of Innate Immune Responses by Autophagy: A Goldmine for Viruses** *Front Immunol* **11** <https://doi.org/10.3389/fimmu.2020.578038>
- Puvirajesinghe T. M., Bertucci F., Jain A., Scerbo P., Belotti E., Audebert S., Borg J. P (2016) **Identification of p62/SQSTM1 as a component of non-canonical Wnt VANGL2-JNK signalling in breast cancer** *Nat Commun* **7** <https://doi.org/10.1038/ncomms10318>
- Shaid S., Brandts C. H., Serve H., Dikic I (2013) **Ubiquitination and selective autophagy** *Cell Death Differ* **20**:21–30 <https://doi.org/10.1038/cdd.2012.72>

Tan P., He L., Xing C., Mao J., Yu X., Zhu M., Wang R. F (2019) **Myeloid loss of Beclin 1 promotes PD-L1hi precursor B cell lymphoma development** *J Clin Invest* **129**:5261–5277 <https://doi.org/10.1172/JCI127721>

Tong Y., Cui J., Li Q., Zou J., Wang H. Y., Wang R. F (2012) **Enhanced TLR-induced NF-kappaB signaling and type I interferon responses in NLRC5 deficient mice** *Cell Res* **22**:822–835 <https://doi.org/10.1038/cr.2012.53>

Wu Y., Jin S., Liu Q., Zhang Y., Ma L., Zhao Z., Cui J (2021) **Selective autophagy controls the stability of transcription factor IRF3 to balance type I interferon production and immune suppression** *Autophagy* **17**:1379–1392 <https://doi.org/10.1080/15548627.2020.1761653>

Yamano K., Youle R. J (2020) **Two different axes CALCOCO2-RB1CC1 and OPTN-ATG9A initiate PRKN-mediated mitophagy** *Autophagy* **16**:2105–2107 <https://doi.org/10.1080/15548627.2020.1815457>

Yang L., Zhang T., Zhang C., Xiao C., Bai X., Wang G (2021) **Upregulated E3 ligase tripartite motif-containing protein 21 in psoriatic epidermis ubiquitylates nuclear factor-kappaB p65 subunit and promotes inflammation in keratinocytes** *Br J Dermatol* **184**:111–122 <https://doi.org/10.1111/bjd.19057>

Yin Z., Popelka H., Lei Y., Yang Y., Klionsky D. J (2020) **The Roles of Ubiquitin in Mediating Autophagy** *Cells* **9** <https://doi.org/10.3390/cells9092025>

Editors

Reviewing Editor

Soumen Basak

National Institute of Immunology, New Delhi, India

Senior Editor

Satyajit Rath

Indian Institute of Science Education and Research (IISER), Pune, India

Reviewer #1 (Public Review):

The study shows a new mechanism of NFkB-p65 regulation mediated by Vangl2-dependent autophagic targeting. Autophagic regulation of p65 has been reported earlier; this study brings an additional set of molecular players involved in this important regulatory event, which may have implications for chronic and acute inflammatory conditions.

Comments on the revised version:

The authors have addressed the earlier concerns and I am satisfied with the revised version. I have no additional comments to make.

<https://doi.org/10.7554/eLife.87935.2.sa2>

Reviewer #2 (Public Review):

Vangl2, a core planar cell polarity protein involved in Wnt/PCP signaling, cell proliferation, differentiation, homeostasis, and cell migration. Vangl2 malfunctioning has been linked to

various human ailments, including autoimmune and neoplastic disorders. Interestingly, it was shown that Vangl2 interacts with the autophagy regulator p62, and autophagic degradation limits the activity of inflammatory mediators, such as p65/NF- κ B. However, the possible role of Vangl2 in inflammation has not been investigated. In this manuscript, Lu et al. describe that Vangl2 expression is upregulated in human sepsis-associated PBMCs and that Vangl2 mitigates experimental sepsis in mice by negatively regulating p65/NF- κ B signaling in myeloid cells. Their mechanistic studies further revealed that Vangl2 recruits the E3 ubiquitin ligase PDLIM2 to promote K63-linked poly-ubiquitination of p65. Vangl2 also facilitated the recognition of ubiquitinated p65 by the cargo receptor NDP52. These molecular processes caused selective autophagic degradation of p65. Indeed, abrogation of PDLIM2 or NDP52 functions rescued p65 from autophagic degradation, leading to extended p65/NF- κ B activity in myeloid cells. Overall, the manuscript presents convincing evidence for novel Vangl2-mediated control of inflammatory p65/NF- κ B activity. The proposed pathway may expand interventional opportunities restraining aberrant p65/NF- κ B activity in human ailments.

IKK is known to mediate p65 phosphorylation, which instructs NF- κ B transcriptional activity. In this manuscript, Vangl2 deficiency led to an increased accumulation of phosphorylated p65 and IKK also at 30 minutes post-LPS stimulation; however, autophagic degradation of p-p65 may not have been initiated at this early time point. Therefore, this set of data put forward the exciting possibility that Vangl2 could also be regulating the immediate early phase of inflammatory response involving the IKK-p65 axis - a proposition that may be tested in future studies.

<https://doi.org/10.7554/eLife.87935.2.sa1>

Reviewer #3 (Public Review):

Lu et al. describe Vangl2 as a negative regulator of inflammation in myeloid cells. The primary mechanism appears to be through binding p65 and promoting its degradation, albeit in an unusual autolysosome/autophagy dependent manner. Overall, these findings are novel, valuable and the crosstalk of PCP pathway protein Vangl2 with NF- κ B is of interest. While generally solid, some concerns still remain about the rigor and conclusions drawn.

Comments on the revised version:

Lu et al. address my comments through responses and new experimental data. However, some of the explanations provided are inadequate.

The new experimental data using phosphomutants indeed adds to their claim that this is a PCP-independent function of Vangl2.

The addition of statistics and testing JNK pathway is appreciated by this Reviewer.

However, in response to my enquiry regarding directly exploring PCP effects, the authors simply assert "Our study revealed that Vangl2 recruits the E3 ubiquitin ligase PDLIM2 to facilitate K63-linked ubiquitination of p65, which is subsequently recognized by autophagy receptor NDP52 and then promotes the autophagic degradation of p65. Our findings by using autophagy inhibitors and autophagy-deficient cells indicate that Vangl2 regulates NF κ B signaling through a selective autophagic pathway, rather than affecting the PCP pathway, WNT, HH/GLI, Fat-Dachsous or even mechanical tension."

I do not agree that the use of autophagy inhibitors and autophagy-deficient cells can rule out the contributions of PCP or any other pathways. Only experimentally inhibiting the pathway(s) with adequate demonstration of target inhibition/abolition of well-known effector function and documenting unaltered p65 regulation under these conditions can be considered proof. Autophagy inhibitors and autophagy-deficient cells only prove that this

particular pathway is necessary. Nonetheless, I do not want to dwell on proving a negative and agree that Vangl2 is a novel regulator of p65 through its role in promoting p65 degradation. The inclusion of a statement discussing the limitations of their approach would have sufficed. The response from the authors could have been better.

I am also not satisfied with the explanation that "immune cells represent a minor fraction of the lungs and liver". There are lots of resident immune cells in the lungs and liver (alveolar macrophages in the lung and Kupffer cells in the liver). For example, it may be so that Vangl2 is important in monocytes and not in the resident population. This might be a potential explanation. But this is not explored. The restricted tissue-specificity of the interaction between two ubiquitously present proteins is still a challenge to understand. The response from the authors is not satisfactory. There is plenty of Vangl2 in the liver in their western blot.

I had also simply pointed out PMID: 34214490 with reference to the findings described in the manuscript. There were no suggestions of contradiction. In fact, I would refer to the publication in discussion to support the findings and stress the novelty. The response from the authors could have been better.

The response to my enquiry regarding homo- or heterozygosity is unsupported by any reference or data.

The listing of 8 patients and healthy controls are also appreciated. The body temperature of #6 doesn't fall in the <36 or >38 degree C SIRS criteria. The inclusion of CRP, PCT, heart rate and respiratory rate, and other lab values would have further improved the inclusion criteria. Moreover, it is difficult to understand why there are 16 value points for healthy and sepsis cohorts in Fig 1 when there are 8 patients.

<https://doi.org/10.7554/eLife.87935.2.sa0>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

In the manuscript titled "Vangl2 suppresses NF- κ B signaling and ameliorates sepsis by targeting p65 for NDP52-mediated autophagic degradation" by Lu et al, the authors show that Vangl2, a planner cell polarity component, plays a direct role in autophagic degradation of NF κ B-p65 by facilitating its ubiquitination via PDLIM2 and subsequent recognition and autophagic targeting via the autophagy adaptor protein NDP52. Conceptually it is a wonderful study with excellent execution of experiments and controls. The concerns with the manuscript are mainly on two counts - First issue is the kinetics of p65 regulation reported here, which does not fit into the kinetics of the mechanism proposed here, i.e., Vangl2-mediated ubiquitination followed by autophagic degradation of p65. The second issue is more technical- an absolute lack of quantitative analyses. The authors rely mostly on visual qualitative interpretation to assess an increase or decrease in associations between partner molecules throughout the study. While the overall mechanism is interesting, the authors should address these concerns as highlighted below:

Major points:

(1) Kinetics of p65 regulation by Vangl2: As mentioned above, authors report that LPS stimulation leads to higher IKK and p65 activation in the absence of Vangl2. The mechanism of action authors subsequently work out is that- Vangl2 helps recruit E3 ligase PDLIM to p65, which causes K63 ubiquitination, which is recognised by NDP52 for autophagic targeting. Curiously, peak p65 activation is achieved within 30 minutes of LPS stimulation. The time scale of all other assays is way longer. It is not clear that in WT cells, p65 could be targeted to autophagic degradation in Vangl2 dependent manner within 30 minutes. The HA-Myc-Flag-based overexpression and Co-IP studies do confirm the interactions as proposed. However, they do not prove that this mechanism was responsible for the Vangl2-mediated modulation of p65 activation upon LPS stimulation. Moreover, the Vangl2 KO line also shows increased IKK activation. The authors do not show the cause behind increased IKK activation, which in itself can trigger increased p65 phosphorylation.

We thank the reviewer for this valuable suggestion.

Indeed, we agreed with the reviewer that peak p65 activation is achieved within 30 minutes of LPS stimulation in vitro, and p65 could not be targeted to autophagic degradation in a Vangl2 dependent manner within 30 minutes. Given that the protein and mRNA levels of Vangl2 were elevated at 3-6 h of LPS stimulation (Fig. S1 C-E), we extended the stimulation time scale in the revised manuscript. The data (Fig. 2A-D in the revised manuscript) demonstrated that IKK phosphorylation was enhanced in Vangl2 KO myeloid cells during the early phase (within 3 h) of LPS stimulation, but not for the prolonged period of LPS stimulation. The underlying mechanism may be complex. Only p65 phosphorylation was continuously enhanced after long-term LPS stimulation in Vangl2 KO cells, compared to WT cells. Furthermore, the overexpression of Vangl2 in A549 cells also demonstrated a reduction of phosphorylation and total endogenous p65 (Fig. 2 I, J in the revised manuscript). These findings were corroborated by overexpression and Co-IP experiments, which collectively indicated that Vangl2 regulates the stability of p65 by promoting its interaction with NDP52 and autophagic degradation. (Page 7; Line 183-185).

(2) The other major concern is regarding the lack of quantitative assessments. For Co-IP experiments, I can understand it is qualitative observation. However, when the authors infer that there is an increase or decrease in the association through co-IP immunoblots, it should also be quantified, especially since the differences are quite marginal and could be easily misinterpreted.

We are grateful to the reviewer for this suggestion. The quantitative analysis has been updated in the revised version.

(3) Figure 4E and F: It is evident that inhibiting Autolysosome (CQ or BafA1) or autophagy (3MA) led to the recovery of p65 levels and inducing autophagy by Rapamycin led to faster decay in p65 levels. Did the authors also note/explore the possibility that Vangl2 itself may be degraded via the autophagy pathway? IB of WCL upon CQ/BAF/3MA or upon Rapa treatment does indicate the same. If true, how would that impact the dynamics of p65 activation?

We thank the reviewer for this question. Previous studies have shown that Vangl2 is primarily degraded by the proteasome pathway, rather than by the autolysosomal pathway (doi: 10.1126/sciadv.abg2099; doi: 10.1038/s41598-019-39642-z). In our experiments, Vangl2 recruits E3 ligase PDLIM2 to enhance K63-linked ubiquitination on p65, which serves as a recognition signal for cargo receptor NDP52-mediated selective autophagic degradation. Vangl2 facilitated the interaction between p65 and NDP52, yet itself did not undergo significant autophagic degradation.

(4) Autophagic targeting of p65 should also be shown through alternate evidence, like microscopy etc., in the LPS-stimulated WT cells.

We thank the reviewer for this suggestion. We have added the data (co-localization of p65 and LC3 was detected by immunofluorescence) in the revised version (Fig. S4 H in the revised manuscript). (Page 9, lines 267-268)

Reviewer #2 (Public Review):

Vangl2, a core planar cell polarity protein involved in Wnt/PCP signaling, mediates cell proliferation, differentiation, homeostasis, and cell migration. Vangl2 malfunctioning has been linked to various human ailments, including autoimmune and neoplastic disorders. Interestingly, Vangl2 was shown to interact with the autophagy regulator p62, and indeed, autophagic degradation limits the activity of inflammatory mediators such as p65/NF- κ B. However, if Vangl2, per se, contributes to restraining aberrant p65/NF- κ B activity remains unclear.

In this manuscript, Lu et al. describe that Vangl2 expression is upregulated in human sepsis-associated PBMCs and that Vangl2 mitigates experimental sepsis in mice by negatively regulating p65/NF- κ B signaling in myeloid cells. Vangl2 recruits the E3 ubiquitin ligase PDLIM2 to promote K63-linked poly-ubiquitination of p65. Vangl2 also facilitates the recognition of ubiquitinated p65 by the cargo receptor NDP52. These molecular processes cause selective autophagic degradation of p65. Indeed, abrogation of PDLIM2 or NDP52 functions rescued p65 from autophagic degradation, leading to extended p65/NF- κ B activity.

As such, the manuscript presents a substantial body of interesting work and a novel mechanism of NF- κ B control. If found true, the proposed mechanism may expand therapeutic opportunities for inflammatory diseases. However, the current draft has significant weaknesses that need to be addressed.

We appreciate the reviewer's comments on our manuscript, and we have further improved the manuscript as suggested.

Specific comments

(1) Vangl2 deficiency did not cause a discernible increase in the cellular level of total endogenous p65 (Fig 2A and Fig 2B) but accumulated also phosphorylated IKK.

Even Fig 4D reveals that Vangl2 exerts a rather modest effect on the total p65 level and the figure does not provide any standard error for the quantified data. Therefore, these results do not fully support the proposed model (Figure 7) - this is a significant drawback. Instead, these data provoke an alternate hypothesis that Vangl2 could be specifically mediating autophagic removal of phosphorylated IKK and phosphorylated IKK, leading to exacerbated inflammatory NF- κ B response in Vangl2-deficient cells. One may need to use phosphorylation-defective mutants of p65, at least in the over-expression experiments, to dissect between these possibilities.

We appreciate the reviewer's comments on our manuscript, and we have further improved the manuscript as suggested.

(1) Indeed, we agreed with the reviewer that Vangl2 deficiency did not cause a discernible increase in the cellular level of total p65 after a short time of LPS stimulation in vitro, and p65 could not be targeted to autophagic degradation in a Vangl2 dependent manner within 30 minutes. Given that the protein and mRNA levels of Vangl2 were elevated at 3-6 h of LPS

stimulation (Fig. S1 C-E), we extended the stimulation time scale in the revised manuscript. The data (Fig. 2A-D in the revised manuscript) demonstrated that IKK phosphorylation was enhanced in Vangl2 KO myeloid cells during the early phase (within 3 h) of LPS stimulation, but not for the prolonged period of LPS stimulation. The underlying mechanism may be complex. Only phosphorylation of p65 and total endogenous p65 was continuously enhanced after long-term LPS stimulation in Vangl2 KO cells, compared to WT cells. Furthermore, the overexpression of Vangl2 in A549 cells also demonstrated a reduction of phosphorylation and total endogenous p65 (Fig. 2 I, J in the revised manuscript). These findings were corroborated by overexpression and Co-IP experiments, which collectively indicated that Vangl2 regulates the stability of p65 by promoting its interaction with NDP52 and autophagic degradation. (Page 7; Line 183-185).

(2) Similarly, the stimulation time scale in Fig 4D was extended, and it was demonstrated that p65 was more stable in Vangl2-deficient cells.

1. Moreover, we constructed phosphorylation-defective mutants of p65 (S536A), and found that Vangl2 could also promote the degradation of the p65 phosphorylation mutants (Fig. S4 A, B in the revised manuscript). Thus, Vangl2 promote the degradation of the basal/unphosphorylated p65. (Page 8, lines 237-240)

(2) Fig 1A: The data indicates the presence of two subgroups within the sepsis cohort - one with high Vangl2 expressions and the other with relatively normal Vangl2 expression. Was there any difference with respect to NF- κ B target inflammatory gene expressions between these subgroups?

As suggested, we conducted an analysis of NF- κ B target inflammatory gene expressions between the high and relatively low Vangl2 expression groups in sepsis patients. The results showed that the serum of the high Vangl2 expression group exhibited lower levels of IL-6, WBC, and CRP than the low Vangl2 expression group, which suggested an inverse correlation between Vangl2 and the inflammatory response (Fig. S1 A in the revised manuscript) (Page 5, lines 126-128).

(3) The effect of Vangl2 deficiency was rather modest in the neutrophil. Could it be that Vangl2 mediates its effect mostly in macrophages?

As showed in Fig. S1C-E, the induction of Vangl2 by LPS stimulation is more rapid in macrophages than in neutrophils. This may contribute to its dominant effect in macrophages. Consequently, we primarily focused our investigation on the role of Vangl2 in macrophages.

(4) Fig 1D and Figure 1E: Data for unstimulated Vangl2 cells should be provided. Also, the source of the IL-1 β primary antibody has not been mentioned.

Thank you for the suggestion. We have updated the data for unstimulated cells in the revised manuscript (Fig. 1 D, E in the revised manuscript). Also, IL-1 β primary antibody was purchased from Cell Signaling Technology and the information has been included in the Materials and Methods section (Table S1).

(5) The relevance and the requirement of RNA-seq analysis are not clear in the present draft. Figure 1E already reveals upregulation of the signature NF- κ B target inflammatory genes upon Vangl2 deficiency.

We agreed with the reviewer that the data presented in Figure 1E demonstrated the upregulation of the signature NF- κ B target inflammatory genes upon Vangl2 deficiency in a murine model of LPS induced sepsis. Subsequently, we proceeded to investigate the

mechanism by which Vangl2 regulates NF- κ B target inflammatory genes at the cellular level in Figure 2. To this end, we performed RNA-seq analysis to screen signal pathways involved in LPS-induced septic shock by comparing LPS-stimulated BMDMs from Vangl2 Δ M and WT mice, and identified that TNF signaling pathway and cytokine-cytokine receptor interaction were found to be significantly enriched in Vangl2 Δ M BMDMs upon LPS stimulation. This analysis provides further evidence that Vangl2 plays a role in regulating NF- κ B signaling pathways and the release of related inflammatory cytokines.

(6) Fig 2A reveals an increased accumulation of phosphorylated p65 and IKK in Vangl2-deficient macrophages upon LPS stimulation within 30 minutes. However, Vangl2 accumulates at around 60 minutes post-stimulation in WT cells. Similar results were obtained for neutrophils (Fig 2B). There appears to be a temporal disconnect between Vangl2 and phosphorylated p65 accumulation - this must be clarified.

This concern has been addressed above (see response to questions 1 from reviewer #2).

(7) Figure 2E and 2F do not have untreated controls. Presentations in Fig 2E may be improved to more clearly depict IL6 and TNF data, preferably with separate Y-axes.

Thank you for the suggestion. We have added untreated controls and separated Y-axes for IL-6 and TNF data in the revised manuscript (Fig. 2 E, F in the revised manuscript).

(8) Line 219: "strongly with IKK α , p65 and MyD88, and weak" - should be revised.

We have improved the manuscript as suggested in the revised manuscript (Page 7; Line 203).

(9) It is not clear why IKK β was excluded from interaction studies in Fig S3G.

We added the Co-IP experiment and showed that HA-tagged Vangl2 only interacted with Flag-tagged p65, but not with Flag-tagged IKK β in 293T cells (Fig S3H). Furthermore, endogenous co-IP immunoblot analyses showed that Vangl2 did not associate with IKK β (Fig. S3I)

(10) Fig 3F- In the text, authors mentioned that Vangl2 strongly associates with p65 upon LPS stimulation in BMDM. However, no controls, including input or another p65-interacting protein, were used.

As reviewer suggested, we have added input and positive control (IkBa) in this experiment (Fig. 3F in the revised manuscript). The results demonstrated that the interaction between p65 and IkBa was attenuated, although the total IkBa did not undergo significant degradation over long-term course of LPS stimulation.

(11) Figure 4D - Authors claim that Vangl2-deficient BMDMs stabilized the expression of endogenous p65 after LPS treatment. However, p65 levels were particularly constitutively elevated in knockout cells, and LPS signaling did not cause any further upregulation. This again indicates the role of Vangl2 in the basal state. The authors need to explain this and revise the test accordingly.

Thank you for the reviewer's comments. We repeated the experiment to ascertain whether Vangl2 could stabilize the expression of endogenous p65 before and after LPS treatment. It was found that, due to the extremely low expression of Vangl2 in WT cells in the absence of stimulation, there was no observable difference on the basal level of p65 between WT and Vangl2 Δ M cells. However, upon prolonged LPS stimulation, Vangl2 expression was induced, resulting in p65 degradation in WT cells. In contrast, p65 protein was more stable in Vangl2 deficient cells after LPS stimulation (Fig. 4D in the revised manuscript).

Reviewer #3 (Public Review):

Lu et al. describe Vangl2 as a negative regulator of inflammation in myeloid cells. The primary mechanism appears to be through binding p65 and promoting its degradation, albeit in an unusual autolysosome/autophagy dependent manner. Overall, the findings are novel and the crosstalk of PCP pathway protein Vangl2 with NF-kappaB is of interest.Regardless, Vangl2 as a negative regulator of NF-kappaB is an important finding. There are, however, some concerns about methodology and statistics that need to be addressed.

Thank you for your comments on our manuscript, and we have further improved the manuscript as suggested.

(1) Whether PCP is anyway relevant or if this is a PCP-independent function of Vangl2 is not directly explored (the later appears more likely from the manuscript/discussion). PCP pathways intersect often with developmentally important pathways such as WNT, HH/GLI, Fat-Dachsous and even mechanical tension. It might be of importance to investigate whether Vangl2-dependent NF-kappaB is influenced by developmental pathways.

Thank you for the reviewer's insightful comments. Our study revealed that Vangl2 recruits the E3 ubiquitin ligase PDLIM2 to facilitate K63-linked ubiquitination of p65, which is subsequently recognized by autophagy receptor NDP52 and then promotes the autophagic degradation of p65. Our findings by using autophagy inhibitors and autophagic-deficient cells indicate that Vangl2 regulates NF-kB signaling through a selective autophagic pathway, rather than affecting the PCP pathway, WNT, HH/GLI, Fat-Dachsous or even mechanical tension. Moreover, a discussion section has been added to the revised version. (Page 12, lines 377-393)

(2) Are Vangl2 phosphorylations (S5, S82 and S84) in anyway necessary for the observed effects on NF-kappaB or would a phospho-mutant (alanine substitution mutant) Vangl2 phenocopy WT Vangl2 for regulation of NF-kappaB?

As suggested, we generated phospho-mutants of Vangl2 (S82/84A) and observed that Vangl2 (S82/84A) could still facilitate the degradation of p65 (Fig. S4 B in the revised manuscript), suggesting that Vangl2 regulates the NF-kB pathway independently of its phosphorylation.

(3) Another area to strengthen might be with regards to specificity of cell types where this phenomenon may be observed. LPS treatment in mice resulted in Vangl2 upregulation in spleen and lymph nodes, but not in lung and liver. What explains the specificity of organ/cell-type Vangl2 upregulation and its consequences observed here? Why is NF-kappaB signaling not more broadly or even ubiquitously affected in all cell types in a Vangl2-dependent manner, rather than being restricted to macrophages, neutrophils and peritoneal macrophages, or, for that matter, in spleen and LN and not liver and lung? After all, one may think that the PCP proteins, as well as NF-kappaB, are ubiquitous.

Thank you for the reviewer's comments.

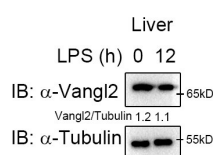
(1) LPS is an important mediator to trigger sepsis with excessive immune activation. As is well known, the spleen and lymph nodes are important peripheral immune organs, where immune cells (e.g., macrophages) are abundant and respond sensitively to LPS stimulation. Nevertheless, immune cells represent a minor fraction of the lungs and liver. Consequently, Vangl2 represents a pivotal regulator of immune function, exhibiting a more pronounced increase in the immune organs and cells.

1. Induction of Vangl2 expression by LPS stimulation is cell specific. Given that different cells exhibit varying protein abundances, the molecular events involved may also differ. Moreover, we observed high Vangl2 expression in the liver at the basal state (Author response image 1), whereas it was not induced after 12 h of LPS stimulation. Therefore, the functional role of Vangl2 exhibits significant phenotype in macrophages and neutrophils/spleen and LN, rather than in liver or lung cells.

Author response image 1.

Vangl2 showed no significant changes in the liver after LPS treatment.

Mice (n≥3) were treated with LPS (30 mg/kg, i.p.). Livers were collected at 12 h after LPS treatment. Immunoblot analysis of Vangl2.



Recommendations For The Authors:

Reviewer #1 (Recommendations For The Authors):

General points:

Figure 4G- panels appear mislabeled. PI correct.

We have corrected this mislabeling as you suggested.

The dynamics of Vangl2 interaction with p65 and autophagy adaptors is not clear/apparent. For example, Vangl2 expression destabilises p65 levels (as in Fig. 4), but in Fig. 5, it seems there is no decline in the p65 protein level, and a large fraction of it coprecipitates with NDP52.

We appreciate the reviewer's comments. In the co-IP assay, we used the lysosomal inhibitor CQ to inhibit p65 degradation to observe the interaction between p65 and NDP52 or Vangl2.

Fig 5E- I would expect p65 levels to be lower in WT cells than Vangl2 KO cells. But as such, there is no difference between the two.

We appreciate the reviewer's comments. We repeated the experiments and updated the data. Firstly, Vangl2 was not induced in WT cells in the absence of LPS stimulation, thus there was no difference in p65 expression between the two groups at the basal level. Secondly, we used CQ/Baf-A1 to inhibit the degradation of Vangl2 in the co-IP assay to observe the interaction between p65 and other molecule.

Reviewer #2 (Recommendations For The Authors):

A few points that can be looked at and revised.

(1) Quantification of the presented data is needed for Fig 4D and Fig 4E.

We added the quantification analysis as suggested.

| (2) *The labeling of Fig 4G should be scrutinized.*

We have corrected this mislabeling as you suggested.

| (3) *Fig 6B and Fig 6C should be explained in the result section more elaborately.*

We thank the reviewer for the suggestion, and we have rephrased this sentence to better describe the results. (Page 10, lines 306-313)

| (4) *Line 85: "Vangl2 mediated downstream of Toll-like or interleukin (IL)-1" - unclear.*

We appreciate the reviewer's comments on our manuscript, and we have further improved the manuscript as suggested in the revised manuscript. (Page 3, lines 68)

| (5) *Line 181: "mice. Differentially expression analysis" - this should be revised.*

We appreciate the reviewer's comments on our manuscript, and we have further improved the manuscript as suggested in the revised manuscript. (Page 11, lines 323)

| (6) *Line 261-264- CHX-chase assay showed the degradation rate of p65 in Vangl2-deficient BMDM was slower compared with WT cells. However, Vangl2 is not induced in WT BMDMs upon CHX treatment (Fig. S4B).*

We appreciate the reviewer's comments on our manuscript, and we have further improved the manuscript as suggested in the revised manuscript (Fig. S4D).

| (7) *Finally, some editing to provide data only critical for the conclusions could improve the ease of reading.*

We have further improved the manuscript as suggested in the revised manuscript.

Reviewer #3 (Recommendations For The Authors):

Comments (general, please address at least in Discussion. Some experimental data, for example the role, if any, of Vangl2 phosphorylations will be very useful):

(1) It might be interesting to explore whether there are any potential effects of developmental pathways on the observed effect mediated by Vangl2 or if the effects are entirely a PCP-independent function of Vangl2. Please see above public review.

Thank you for the reviewer's insightful comments. Our study revealed that Vangl2 recruits the E3 ubiquitin ligase PDLIM2 to facilitate K63-linked ubiquitination of p65, which is subsequently recognized by autophagy receptor NDP52 and then promotes the autophagic degradation of p65. Our findings by using autophagy inhibitors and autophagic-deficient cells indicate that Vangl2 regulates NF- κ B signaling through a selective autophagic pathway, rather than affecting the PCP pathway, WNT, HH/GLI, Fat-Dachsous or even mechanical tension. Furthermore, we generated phospho-mutants of Vangl2 (S82/84A) and observed that Vangl2 (S82/84A) could still facilitate the degradation of p65 (Fig. S4 B), suggesting that Vangl2 regulates the NF- κ B pathway independently of its phosphorylation. In addition, a discussion section has been added to the revised version. (Page 12, lines 377-393)

| (2) *What explains the specificity of organ/cell-type Vangl2 upregulation and its consequences observed here? Why is NF-kappaB signaling not more broadly or even ubiquitously affected in all cell types in a Vangl2-dependent manner, rather than being*

restricted to macrophages, neutrophils and peritoneal macrophages, or, for that matter, in spleen and LN and not liver and lung? Afterall, one may think that the PCP proteins, as well as NF-kappaB, are ubiquitous.

Thank you for the reviewer's comments. A similar question has been addressed above (refer to the response to question 3 of reviewer 3).

(3) Another specificity-related question that comes to mind is whether the Vangl2 function in autolysosomal/autophagic degradation is restricted to p65 as the exclusive substrate? The cytosolic targeting of p65 as opposed to the more well-known nuclear-targeting is interesting.

Our previous finding demonstrated that Vangl2 inhibits antiviral IFN-I signaling by targeting TBK1 for autophagic degradation (doi: 10.1126/sciadv.adg2339), thereby indicating that p65 is not the sole substrate for Vangl2. However, in the NF- κ B pathway, p65 is a specific substrate for Vangl2. Moreover, our findings indicate that the interaction between Vangl2 and p65 occurs predominantly in the cytoplasm, rather than in the nucleus (Fig. S4 C).

(4) Pharmacological approach is used to tease apart autolysosome versus proteasome pathway. What is the physiological importance of autophagic degradation? It is interesting to note that Vangl2 was already previously implicated in degrading LAMP-2A and increasing chaperon-mediated autophagy (CMA)-lysosome numbers (PMID: 34214490).

Previous literature has demonstrated that Vangl2 can inhibit CMA degradation (PMID: 34214490). However, in our study, we found that Vangl2 can promote the selective autophagic degradation of p65. It is important to note that CMA degradation and selective autophagic degradation are two distinct degradation modes, which is not contradictory.

(5) Are these phenotypes discernable in heterozygotes or only when ablated in homozygosity? Any phenotypes recapitulated in the looptail heterozygote mice?

We found that these phenotypes discernable only in homozygosity.

(6) What is the conservation of the Vangl2 p65-interaction site between Vangl2 and Vangl1? PDLIM2 recruitment between Vangl2 and Vangl1?

We appreciate the reviewer's comments on our manuscript. Previous studies have shown that human Vangl1 and Vangl2 exhibit only 72% identity and exhibit distinct functional properties (doi: 10.1530/ERC-14-0141). Thus, the interaction of Vangl2 with p65 and PDLIM2 recruitment may not necessarily occur in Vangl1.

Comments (specific to experiments and data analyses. Please address the following):

(7) The patient population used in Fig 1 is not described in the Methods. This is a critical omission. Were age, sex etc. controlled for between healthy and disease? How was the diagnosis made? What times during sepsis were the samples collected? As presented, this data is impossible to evaluate and interpret.

We appreciate the reviewer's comments on our manuscript, and we have further improved the manuscript as suggested in the revised supplement materials. (Supplementary information, Page 12, lines 146-147)

(8) In general, the statistical method should be described for each experiment presented in the figures. Comparisons should not be made only at the time point with maximal difference (such as in Fig 1F or Fig 2C, but at all time points using appropriate statistical methods). The sample size should also be included to allow determination appropriateness of parametric or non-parametric tests.

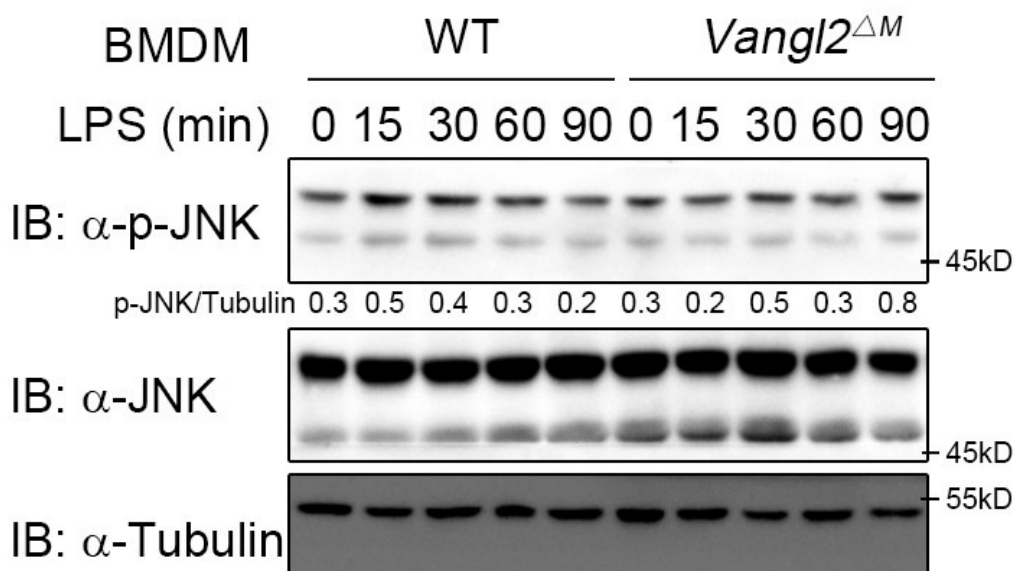
We appreciate the reviewer's comments on our manuscript, and we have further improved the manuscript as suggested in the revised manuscript (Figures 1F and 2C).

(9) PCP pathways can activate p62/SQSTM1 or JNK via RhoA. JNK activation should be tested experimentally.

According to the reviewer's comments, we further examined the effect of Vangl2 on the JNK pathway. The results showed that Vangl2 did not affect the JNK pathway (Author response image 2). This suggests that Vangl2 functions independently of the PCP pathway.

Author response image 1.

Vangl2 did not affect the JNK pathway. WT and Vangl2-deficient ($n \geq 3$) BMDMs were stimulated with LPS (100 ng/ml) for the indicated times. Immunoblot analysis of total and phosphorylated JNK.



(10) Why are different cells such as A549, HEK293, CHO, 293T, THP-1 used during the studies for different experiments? Consistency would improve rigor. At least, logical explanation driving the cell type of choice for each experiment should be included in the manuscript. Nonetheless, one aspect of using a panel of cell lines indicate that the effect of Vangl2 on NF-kappa B is pleiotropic.

We are grateful to the reviewer for their comments on our manuscript. A549, HEK293, CHO, and 293T cells are commonly utilized in protein-protein interaction studies. The selection of cell lines for overexpression (exogenous) experiment is dependent on their transfection

efficiency and the ability to express TLR4 (the receptor for LPS). Additionally, we conducted endogenous experiments by using THP-1 and BMDMs, which are human macrophage cell lines and murine primary macrophages, respectively. Moreover, we generated Vangl2^{f/f} lyz-cre mice by specifically knocking out Vangl2 in myeloid cells, and investigated the effect of Vangl2 on NF- κ B signaling in vivo.

<https://doi.org/10.7554/eLife.87935.2.sa4>